COL774 Assignment 2 Report

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Contents

1 Match result prediction using `Decision Trees`

1.1 Overview

In this section I discuss how I implemented the `Decision Tree`, the `prune()` and the different varations of decision trees we experimented with

1.2 Method of Implementation

1.2.1 The `DecisionTree.fit()`

Each node of the tree is stored as a dictionary containing the following parameters:

- **score**: The Entropy or gini score of the subset of dataset that comes to this node
- **par**: The parent of the node
- **pred**: What to predict if a test example considers this node as leaf
- **depth**: Depth of the node

- **size**: Size of subtree rooted at the node
- **rule**: Column (and median) to split on
- **children**: Dictionary of child nodes indexed by their value of the spliting column
- **id**: An identifier for the node

The Decision tree takes in the names of the categorical and numerical columns as class parameters and these are used to distinguish between categorical and numerical columns.

The tree is built in BFS order using the deque class from collections module.

1.2.2 The `DecisionTree.prune()`

Per-node bookkeeping Each node stores integer arrays and scalars:

`_train_counts`, `_val_counts`, `_test_counts`

and subtree correctness scalars (how many samples in the subtree would be classified correctly if every leaf uses its pred):

`_leaf_corr_train`, `_leaf_corr_val`, `_leaf_corr_test`

Algorithm (high level)

1. **Initialize per-node arrays.** Recursively allocate zero arrays and zero counters for every node.
2. **Route each dataset once.** For each row in each dataset, walk from root down following the node rules. At every visited node increment the appropriate count array at the index of the true class. This fills per-node reach counts for all sets in a single pass per dataset.
3. **Compute leaf-correct subtree counts (postorder).** Do a postorder traversal:
   - If node is a leaf: its `_leaf_corr_*` equals the count for its pred class at that node (i.e. how many samples that actually reach this leaf equal the leaf prediction).
   - If internal: sum its children's `_leaf_corr_*`.

   The root totals give the starting correct counts on each dataset.
4. **Iterative reduced-error pruning:**
   - Build a stable postorder list of nodes (recomputed each pruning iteration).
   - For each internal node (excluding the root), compute the *validation* improvement if the node were collapsed into a leaf predicting the node's current majority training class:
     $$\Delta_{\text{val}} = \texttt{\_val\_counts}[\texttt{pred\_idx}] - \texttt{\_leaf\_corr\_val}.$$
   - Pick the node with largest positive Δ_{val}. If $\max \Delta_{\text{val}} > 0$, physically prune that node:
     (a) Remove children, set `rule` to `None` and `size`=1.
     (b) Update the node's `_leaf_corr_*` to the node's own counts for `pred`.

(c) Compute deltas across train/val/test:

$$\Delta_{\mathrm{tr}} = \texttt{\_train\_counts}[\texttt{pred\_idx}] - \texttt{\_leaf\_corr\_train}$$

(similarly for val/test) and propagate these up the ancestor chain, updating ancestor `_leaf_corr_*` and `size` (subtracting removed subtree size).

- Record new accuracies and tree size, and repeat until no positive validation improvement exists.

Key formulas used

$$\text{accuracy}_{\text{val}} = \frac{\text{root.\_leaf\_corr\_val}}{N_{\text{val}}}$$

When collapsing a node into a leaf with prediction index p:

$$\Delta_{\text{val}} = \text{node.\_val\_counts}[p] - \text{node.\_leaf\_corr\_val}.$$

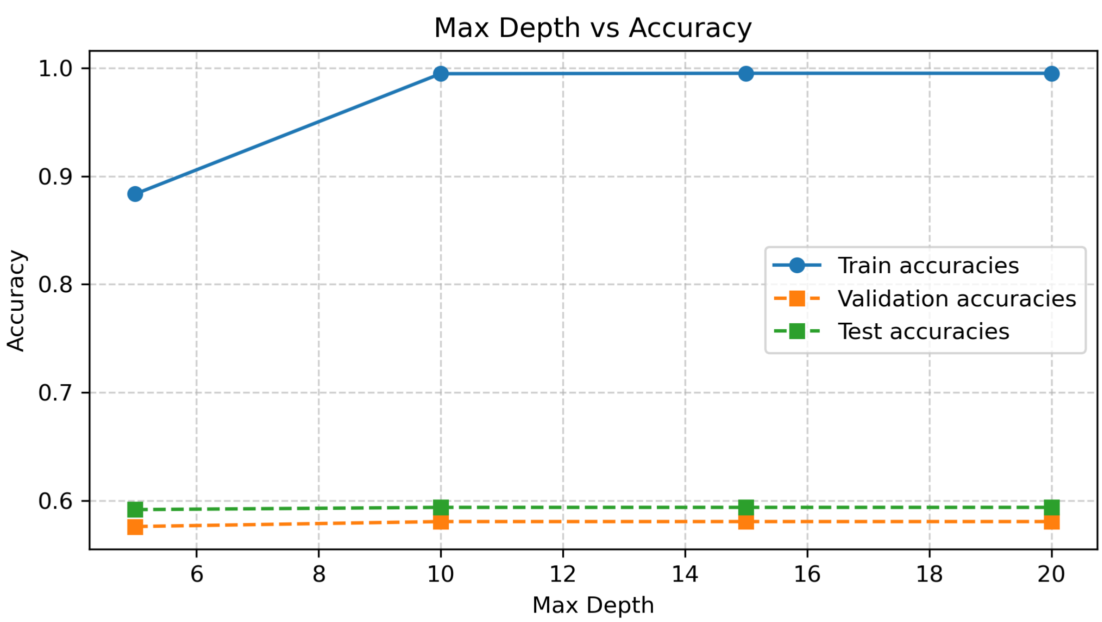
Returned history The function returns lists tracking pruning history:

$$[\, n\_list,\ acc\_train\_list,\ acc\_val\_list,\ acc\_test\_list \,]$$

where *n_list* holds the root size after each prune and the accuracies are recorded after each prune step.

1.2.3 Questions

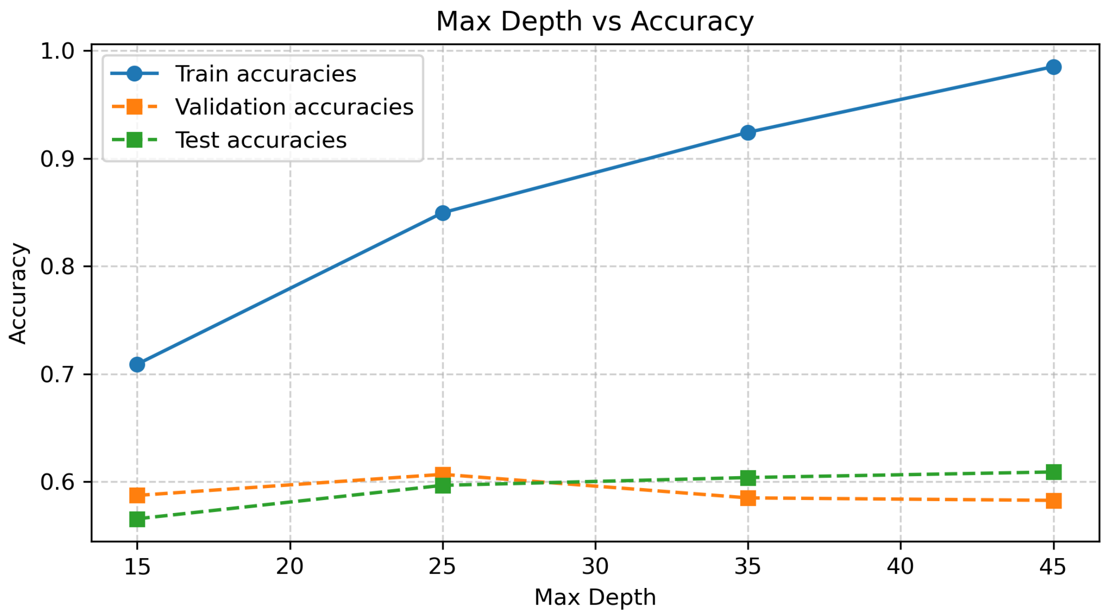
(a) The following results were obtained by using the *max_depth* values $[5, 10, 15, 20]$. Splits in the tree are decided based on the mutual information criterion.



The best model is the one with **max_depth = 10**. The depth of model doesn't change in $max\_depth = [10, 15, 20]$.
The test accuracy is **59.3588%** on the model with **max_depth = 10** which is the best model obtained using the validation data set.

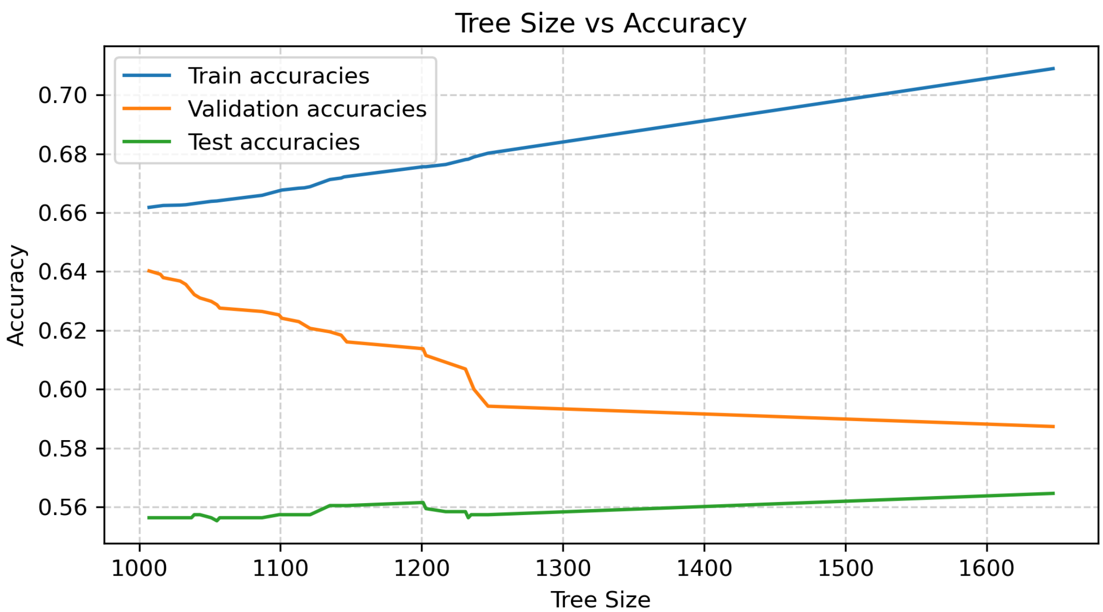
The model is overfitting.

(b) After *one-hot encoding* The following results were obtained by using the max_depth values $[15, 25, 35, 45]$. Splits in the tree are decided based on the mutual information criterion.

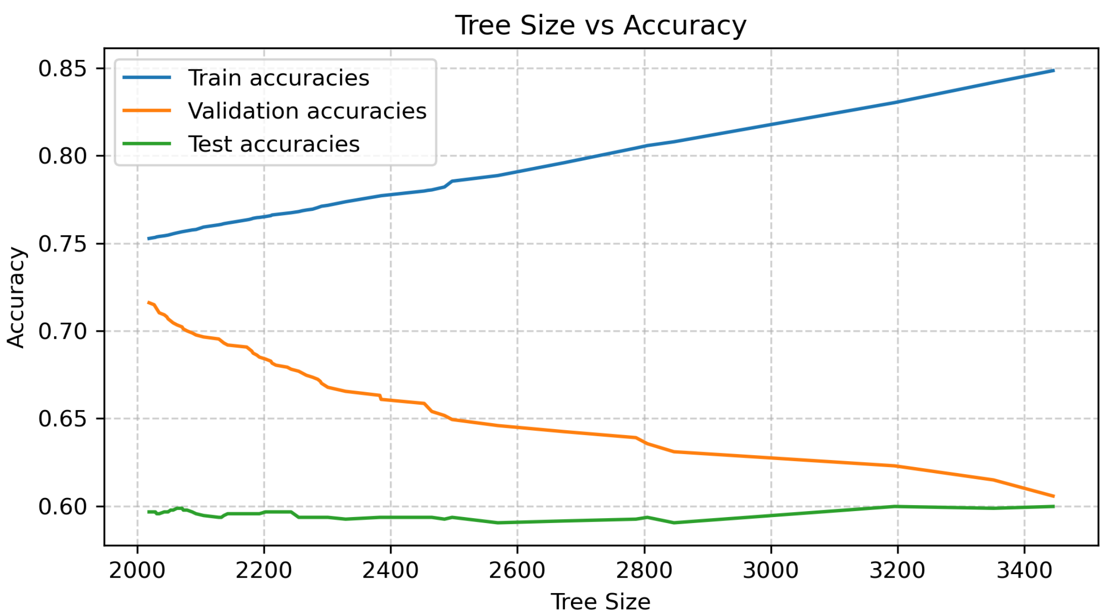


The best model is the one with $max\_depth = 25$. Beyond this, the validation accuracy falls.
The test accuracy is **59.6690%** on the model with **max_depth = 25** which is the best model obtained using the validation data set. Whereas the maximum test accuracy across various depths that was obtained is **60.9100%** obtained at **max_depth = 45**.
This shows a slight improvement from the previous part which was done without one-hot encoding.

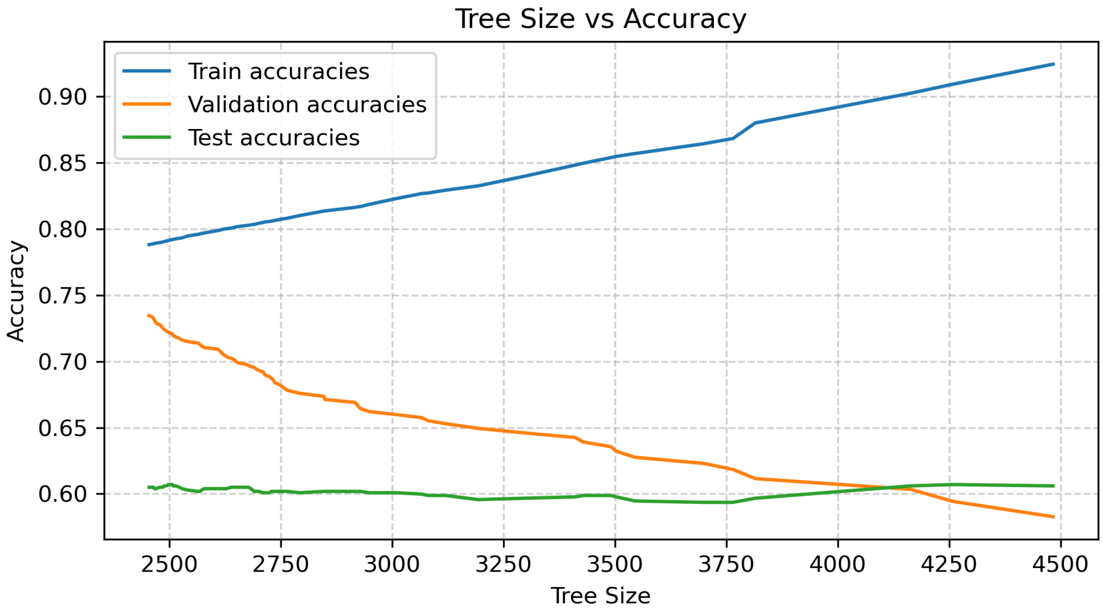
(c) The following plots show the trajectory of the training, validation and test accuracies vs the the number of nodes as we prune the trees for max_depth values form [15, 25, 35, 45]. Splitting was done based on mutual information criterion.



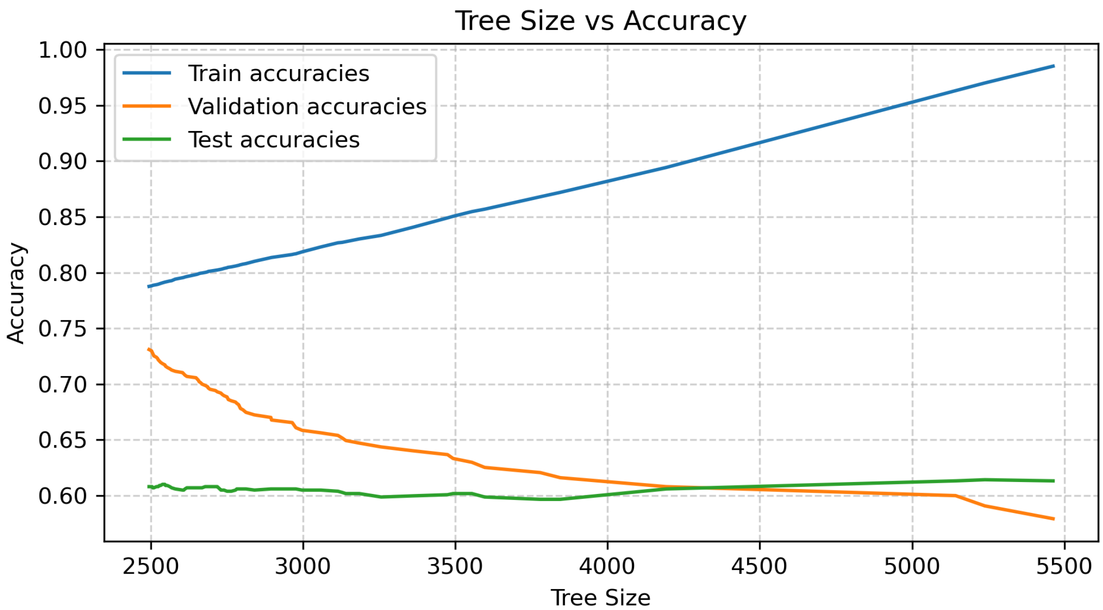
max_depth = 15



max_depth = 25



max_depth = 35



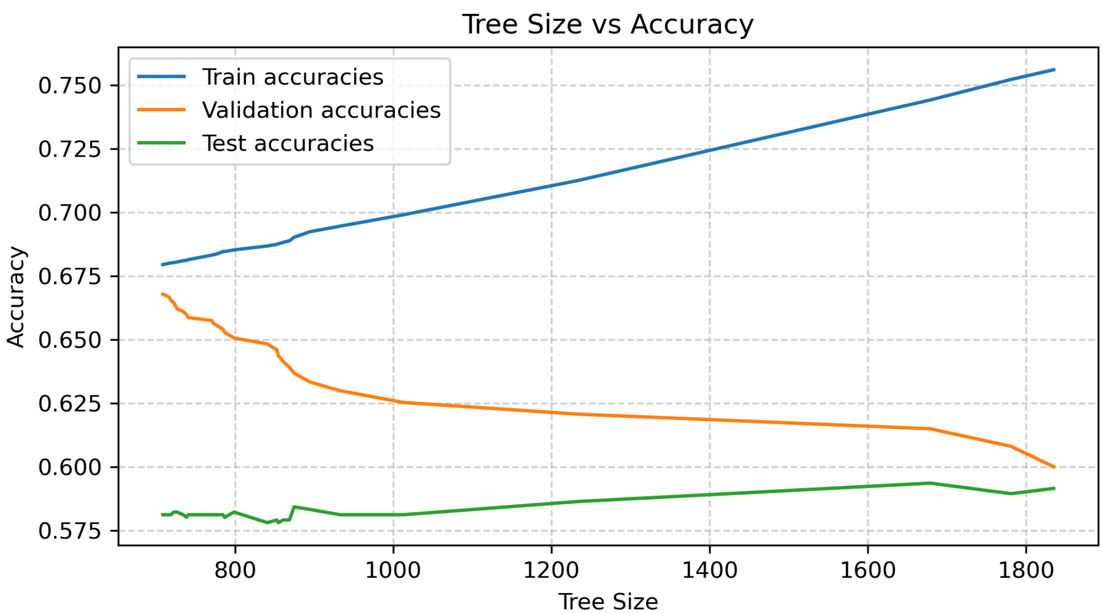
max_depth = 45

The final trees obtained gave the following accuracies on train, validation and test datasets.

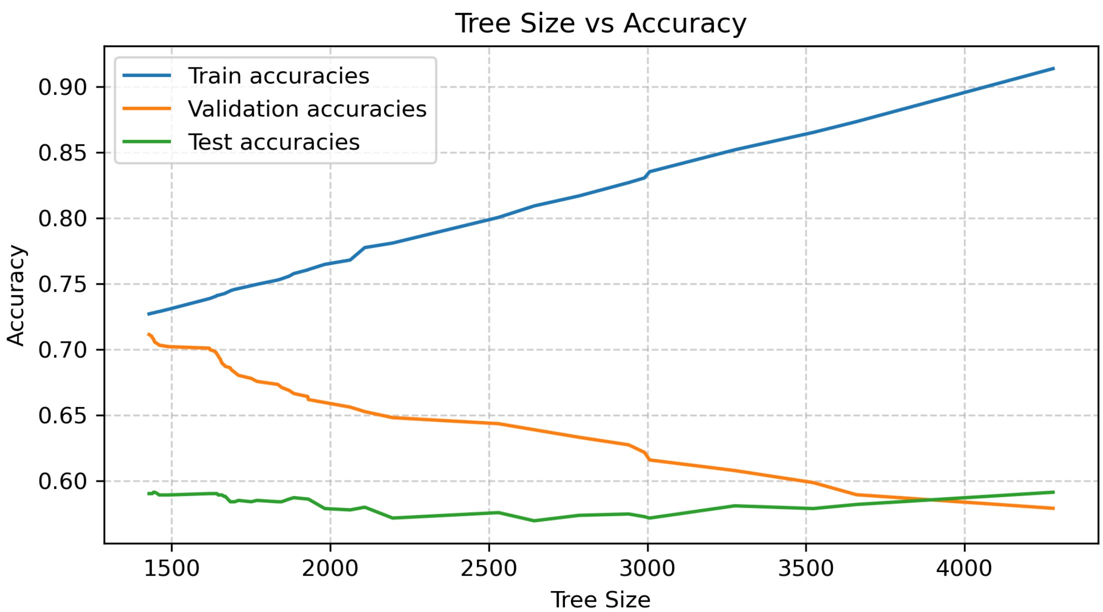
max_depth	Num. Nodes	Train Accuracy	Val Accuracy	Test Accuracy
15	1007	0.6618	0.6402	0.5564
25	2019	0.7528	0.7161	0.5967
35	2455	0.7880	0.7345	0.6050
45	2495	0.7877	0.7310	0.6081

Validation and train accuracies have come close after pruning but for all the cases, test accuracies have not improved. We observe this trend because the given validation dataset is small.
The best tree according to validation accuracies is the tree with **max_depth = 35** and has **2455 nodes**. The test accuracy obtained using this tree is **60.50%**.

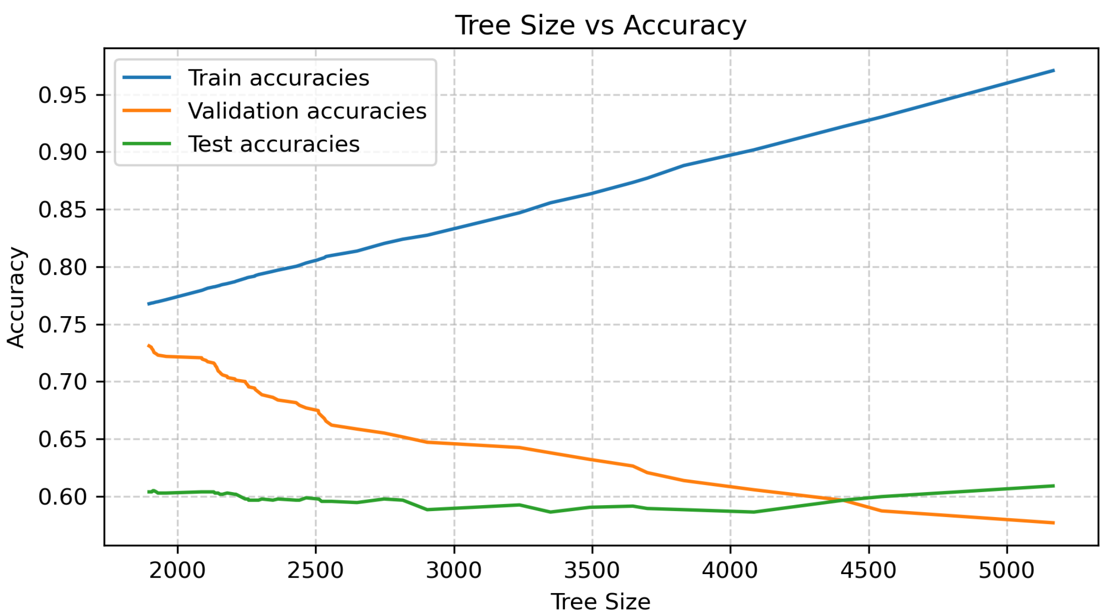
(d) The following plots show the trajectory of the training, validation and test accuracies vs the the number of nodes as we prune the trees for max_depth values form [15, 25, 35, 45]. Splitting was done based on mutual information criterion.



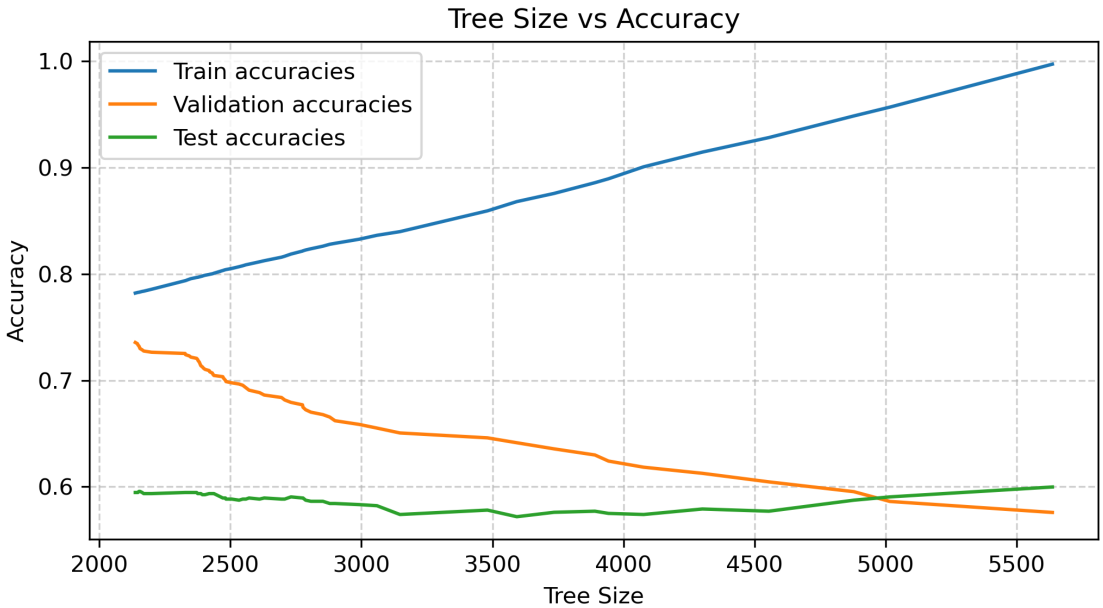
max_depth = 15



max_depth = 25



max_depth = 35



max_depth = 45

The final trees obtained gave the following accuracies on train, validation and test datasets.

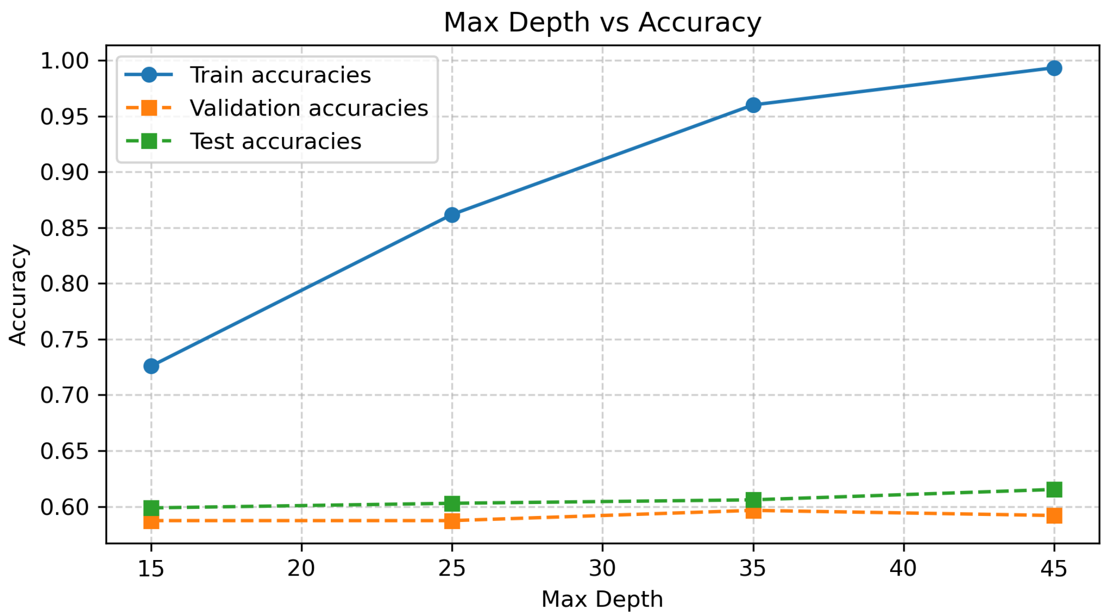
max_depth	Num. Nodes	Train Accuracy	Val Accuracy	Test Accuracy
15	709	0.6794	0.6678	0.5812
25	1429	0.7272	0.7115	0.5905
35	1897	0.7677	0.7310	0.6039
45	2137	0.7820	0.7356	0.5946

Validation and train accuracies have come close after pruning but for all the cases, test accuracies have not improved. We observe this trend because the given validation dataset is small.
The best tree according to validation accuracies is the tree with **max_depth = 45** (35 and 45 trees have similar val accuracies in both the parts (c) and (d)) and has **1897 nodes**. The test accuracy obtained using this tree is **59.46%** whereas the maximum test accuracy is obtained in the tree with **max_depth = 35** i.e. **60.39%**.
The post-pruned trees obtained via gini impurity criterion have lesser number of nodes while maintaining comparable validation and test accuracies, showing effective pruning and reduced model complexity.

(e)(i) In this part sci-kit learn's `DecisionTreeClassifier` was used with splitting criterion as mutual information. The data was one-hot encoded. The following table shows the training, validation and test accuracies obtained by varying the max_depth parameter.

Max Depth	Train Accuracy	Val Accuracy	Test Accuracy
15	0.7258	0.5874	0.5988
25	0.8618	0.5874	0.6029
35	0.9600	0.5966	0.6060
45	0.9934	0.5920	0.6153

The following plot shows the variation in training, validation and test accuracies as max_depth parameter is varied.



The best model is the one with $max\_depth = 35$.
The test accuracy is **60.60%** on the model with **max_depth = 35** which is the best model obtained using the validation data set. Whereas the maximum test accuracy across various depths that was obtained is **61.52%** obtained at **max_depth = 35**.

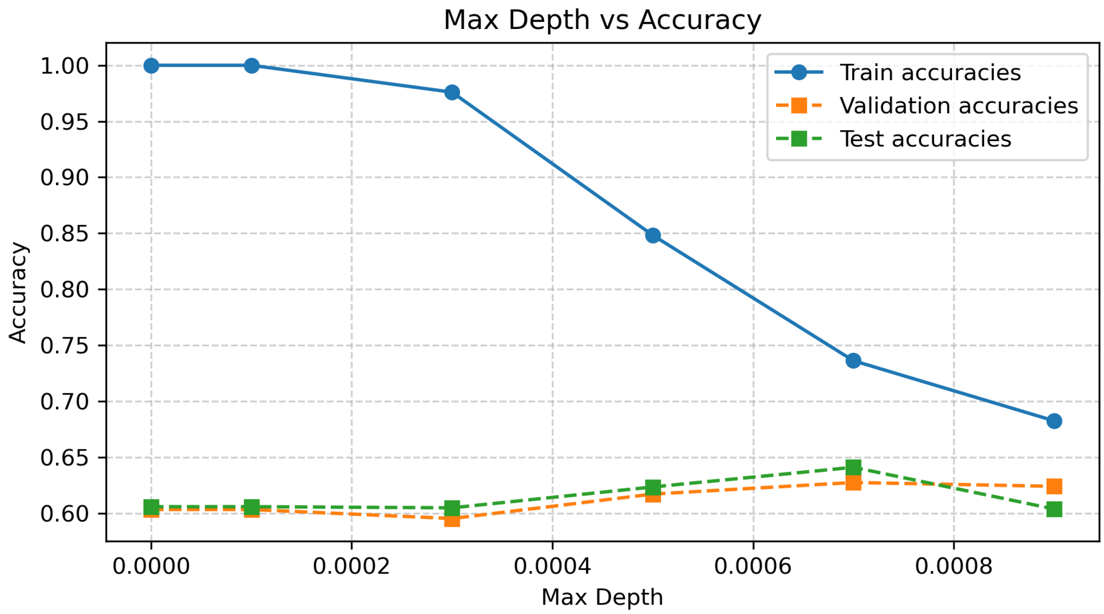
The results obtained here are slightly better than those obtained in part (b)[1.2.3]

(e)(ii) In this part, default value for max_depth parameter was chosen and ccp_alpha parameter was varied using the values from the set $[0.0, 0.0001, 0.0003, 0.0005]$. The following table shows the final accuracies obtained.

Table 1: Effect of Cost-Complexity Pruning (ccp_α) on Accuracy

ccp_α	Train Accuracy	Val Accuracy	Test Accuracy
0.0000	1.0000	0.6034	0.6060
0.0001	1.0000	0.6034	0.6060
0.0003	0.9760	0.5954	0.6050
0.0005	0.8483	0.6172	0.6236
0.0007	0.7363	0.6276	0.6412
0.0009	0.6826	0.6241	0.6039

The following plot shows how the accuracies vary.



ccp_alpha = 0.0005 gives the best validation accuracy and the test accuracy is also the highest seen thus far with a value of **62.25%**. This shows that the *minimal cost-complexity pruning* method is better than *post-pruning* method used in part (c)[2.3]. Also, using higher *ccp_alpha* values gives better results. if the parameter value is **ccp_alpha = 0.0007**, the validation accuracy peaks and test accuracy is **64.12%**.

(f) Random Forest model was trained in this part and the parameters were chosen using GridSearchCV. The following parameter values were given:

```
param_grid =
{ 'n_estimators':[50, 150, 250, 350],
'max_features':[0.1,0.3,0.5,0.7,0.9],
'min_samples_split':[2,4,6,8,10],
'ccp_alpha':[0.0001,0.0003,0.0005,0.0007]
}
```

The best parameters were:

```
{'ccp_alpha': 0.0003, 'max_features': 0.7, 'min_samples_split': 10,
'n_estimators': 150}
```

The following accuracies were obtained:

Training Accuracy: 0.9807
OOB Accuracy: 0.7108
Validation Accuracy: 0.9713
Test Accuracy: 0.6939

The test accuracy obtained here is the highest seen accuracy on the test set thus far. Also, the training and validation accuracies are close. As compared to parts (c) and (d)

the accuracy values obtained here are **significantly higher**, showing that **random forest** algorithm performs overall **better** than simple decision tree with post-pruning.
As the OOB accuracy is close to test accuracy, this shows that the model **generalizes** much **better** than the previous models.

2 Image Classification using Neural Networks

2.1 Overview

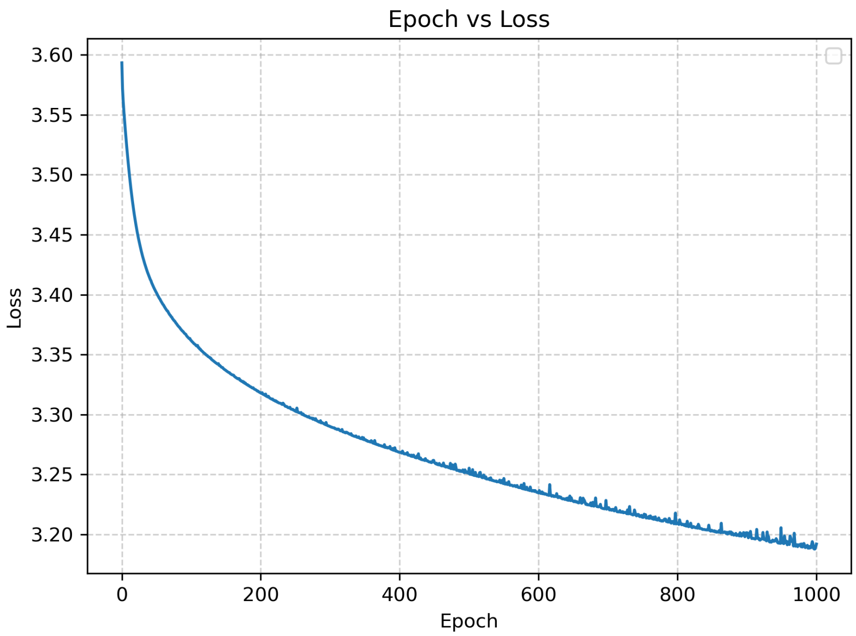
Neural Networks with various architectures was used to identify the given images of consonants of Devanagari script. In this section, I discuss the various architectures of Neural Networks used and the corresponding results obtained.

2.2 Method of Implementation

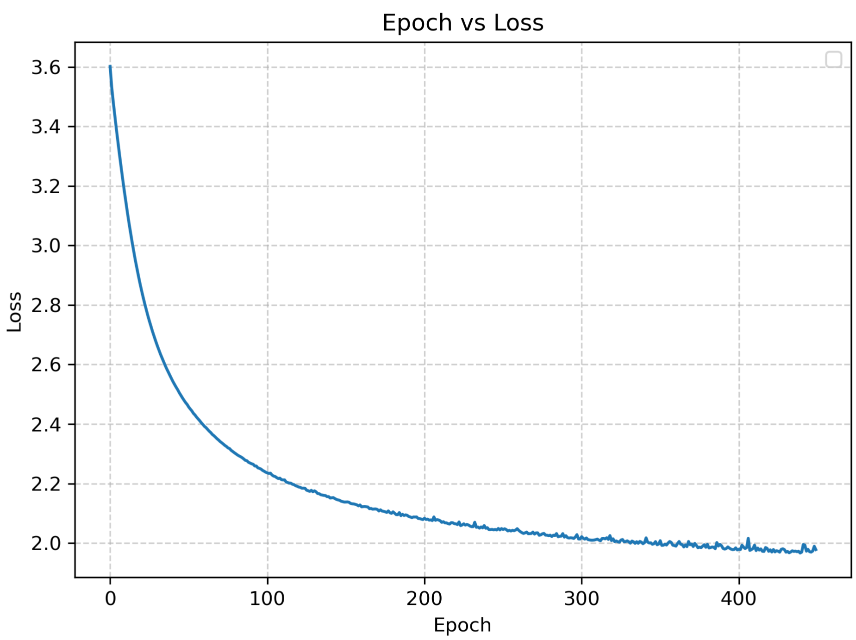
A `NeuralNetwork` class was implemented. It provides flexibility to specify the number of layers and number of units per layer and the activation for each layer. The activation of output layer should be `'softmax'` as we are performing classification task. Cross-entropy loss function was used.
The `NeuralNetwork` class has a `fit()` which takes in the training dataset and the corresponding labels as a one-hot encoded matrix.
The training is done using SGD so, the `fit()` also takes as input *lr*, *batch_size*, *tol*, *n_iter_no_change* and *epochs*. The model is trained for a maximum of *epochs* many steps and the training is stopped prematurely if for *n_iter_no_chage* many steps, the *loss* doesn't reduce atleast by *tol* amount.

2.3 Questions

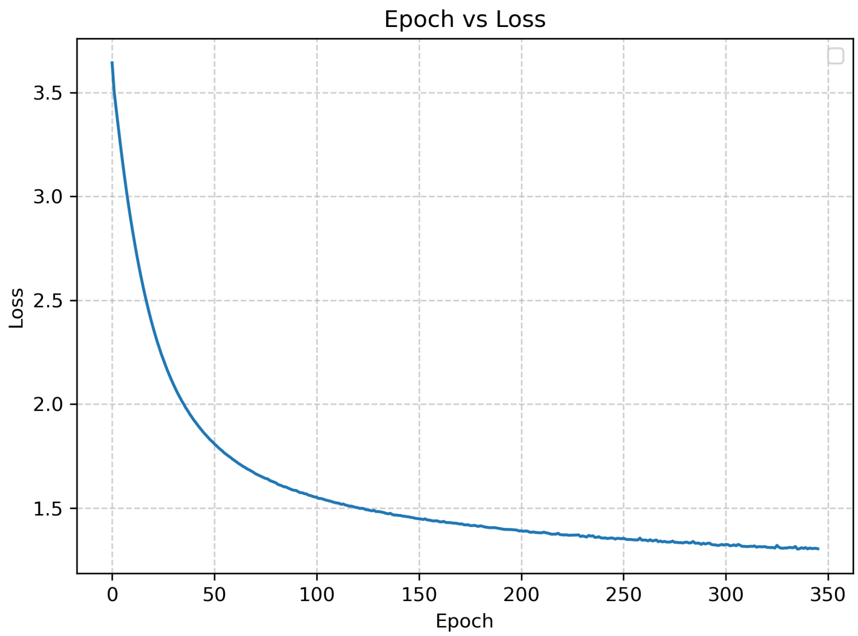
(b) In this question, only one hidden layer was given and the number of units in that layer were take from $[1, 5, 10, 50, 100]$. The models were allowed to train for a max of 1000 epochs due to slow convergence of the model with only 1 unit in the hidden layer. The other parameters were $tol = 1e-5$, $n_iter_no_change = 10$, $lr = 0.01$ and $batch\_size = 32$. The following learning curves were obtained.



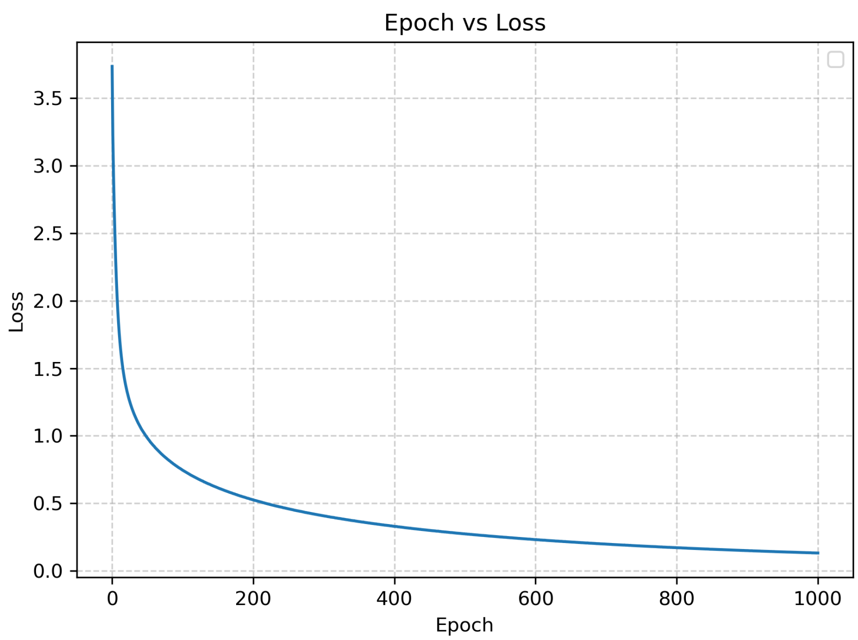
n_units = 1



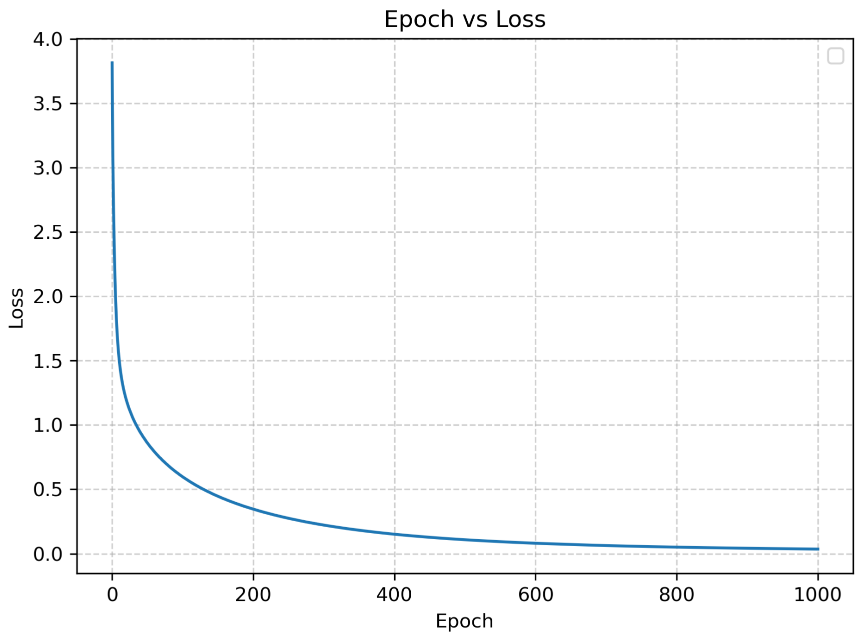
n_units = 5



n_units = 10



n_units = 50



n_units = 100

The following *Classification Reports* were obtained for training data.

Table 2: Classification Report for Training Data ($n\_units = 1$)

Class	Precision	Recall	F1-Score	Support
क	0.00	0.00	0.00	600
ख	0.00	0.00	0.00	600
ग	0.00	0.00	0.00	600
घ	0.00	0.00	0.00	600
ङ	0.04	0.01	0.01	600
च	0.00	0.00	0.00	600
छ	0.00	0.00	0.00	600
ज	0.00	0.00	0.00	600
झ	0.02	0.02	0.02	600
ञ	0.10	0.93	0.17	600
ट	0.05	0.04	0.04	600
ठ	0.05	0.05	0.05	600
ड	0.00	0.00	0.00	600
ढ	0.10	0.97	0.18	600
ण	0.00	0.00	0.00	600
त	0.00	0.00	0.00	600
थ	0.00	0.00	0.00	600
द	0.00	0.00	0.00	600
ध	0.00	0.00	0.00	600

Class	Precision	Recall	F1-Score	Support
न	0.05	0.06	0.06	600
प	0.00	0.00	0.00	600
फ	0.00	0.00	0.00	600
ब	0.00	0.00	0.00	600
भ	0.03	0.02	0.03	600
म	0.06	0.12	0.08	600
य	0.00	0.00	0.00	600
र	0.00	0.00	0.00	600
ल	0.02	0.01	0.01	600
व	0.05	0.05	0.05	600
श	0.05	0.17	0.08	600
ष	0.00	0.00	0.00	600
स	0.05	0.06	0.05	600
ह	0.03	0.01	0.02	600
क्ष	0.06	0.07	0.06	600
त्र	0.00	0.00	0.00	600
ज्ञ	0.06	0.15	0.08	600

Table 3: Classification Report for Training Data ($n\_units = 5$)

Class	Precision	Recall	F1-Score	Support
क	0.61	0.74	0.67	600
ख	0.18	0.06	0.09	600
ग	0.36	0.32	0.34	600
घ	0.34	0.28	0.30	600
ङ	0.33	0.30	0.31	600
च	0.32	0.47	0.38	600
छ	0.48	0.61	0.54	600
ज	0.26	0.39	0.31	600
झ	0.54	0.81	0.65	600
ञ	0.46	0.49	0.48	600
ट	0.60	0.73	0.66	600
ठ	0.56	0.65	0.60	600
ड	0.37	0.36	0.36	600
ढ	0.44	0.77	0.56	600
ण	0.44	0.53	0.48	600
त	0.51	0.42	0.46	600
थ	0.29	0.31	0.30	600
द	0.26	0.15	0.19	600
ध	0.35	0.59	0.44	600
न	0.27	0.32	0.29	600
प	0.40	0.54	0.46	600

Class	Precision	Recall	F1-Score	Support
फ	0.74	0.78	0.76	600
ब	0.29	0.32	0.30	600
भ	0.45	0.56	0.50	600
म	0.26	0.10	0.14	600
य	0.17	0.06	0.09	600
र	0.46	0.41	0.43	600
ल	0.36	0.35	0.35	600
व	0.21	0.10	0.13	600
श	0.28	0.33	0.30	600
ष	0.41	0.42	0.41	600
स	0.19	0.02	0.03	600
ह	0.17	0.04	0.06	600
क्ष	0.33	0.30	0.31	600
त्र	0.26	0.14	0.18	600
ज्ञ	0.36	0.58	0.45	600

Table 4: Classification Report for Training Data ($n\_units = 10$)

Class	Precision	Recall	F1-Score	Support
क	0.80	0.77	0.78	600
ख	0.44	0.39	0.41	600
ग	0.59	0.71	0.64	600
घ	0.51	0.56	0.53	600
ङ	0.53	0.47	0.50	600
च	0.64	0.70	0.67	600
छ	0.64	0.66	0.65	600
ज	0.73	0.65	0.68	600
झ	0.80	0.80	0.80	600
ञ	0.68	0.76	0.72	600
ट	0.76	0.80	0.78	600
ठ	0.76	0.74	0.75	600
ड	0.61	0.64	0.62	600
ढ	0.74	0.76	0.75	600
ण	0.63	0.70	0.66	600
त	0.71	0.74	0.72	600
थ	0.52	0.52	0.52	600
द	0.63	0.55	0.58	600
ध	0.65	0.67	0.66	600
न	0.66	0.56	0.61	600
प	0.54	0.62	0.58	600
फ	0.82	0.84	0.83	600
ब	0.67	0.58	0.62	600

Class	Precision	Recall	F1-Score	Support
भ	0.65	0.68	0.67	600
म	0.57	0.40	0.47	600
य	0.37	0.27	0.31	600
र	0.74	0.80	0.77	600
ल	0.83	0.76	0.79	600
व	0.56	0.52	0.54	600
श	0.49	0.62	0.55	600
ष	0.58	0.72	0.64	600
स	0.49	0.40	0.44	600
ह	0.64	0.53	0.58	600
क्ष	0.55	0.62	0.58	600
त्र	0.60	0.60	0.60	600
ज्ञ	0.66	0.74	0.70	600

Table 5: Classification Report for Training Data ($n\_units = 50$)

Class	Precision	Recall	F1-Score	Support
क	0.99	0.99	0.99	600
ख	0.98	0.99	0.99	600
ग	0.98	0.98	0.98	600
घ	0.99	0.98	0.98	600
ङ	0.97	0.98	0.98	600
च	0.98	0.99	0.99	600
छ	0.98	0.98	0.98	600
ज	0.98	0.98	0.98	600
झ	1.00	1.00	1.00	600
ञ	0.98	0.98	0.98	600
ट	0.99	0.99	0.99	600
ठ	0.99	0.99	0.99	600
ड	0.98	0.97	0.98	600
ढ	0.99	0.99	0.99	600
ण	0.99	1.00	0.99	600
त	0.98	0.99	0.99	600
थ	0.99	0.99	0.99	600
द	0.99	0.99	0.99	600
ध	0.99	0.99	0.99	600
न	0.99	0.98	0.98	600
प	0.98	0.99	0.98	600
फ	0.99	0.99	0.99	600
ब	0.98	0.98	0.98	600
भ	0.99	0.98	0.99	600
म	0.98	0.98	0.98	600

Class	Precision	Recall	F1-Score	Support
य	0.99	0.99	0.99	600
र	0.99	0.99	0.99	600
ल	1.00	0.98	0.99	600
व	0.98	0.96	0.97	600
श	0.98	0.99	0.98	600
ष	0.98	0.98	0.98	600
स	0.99	0.98	0.99	600
ह	1.00	0.99	0.99	600
क्ष	0.99	0.99	0.99	600
त्र	0.99	0.98	0.98	600
ज्ञ	0.99	0.99	0.99	600

Table 6: Classification Report for Training Data ($n\_units = 100$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600

Class	Precision	Recall	F1-Score	Support
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Classification reports for test data

Table 7: Classification Report for Test Data ($n\_units = 1$)

Class	Precision	Recall	F1-Score	Support
क	0.00	0.00	0.00	300
ख	0.00	0.00	0.00	300
ग	0.00	0.00	0.00	300
घ	0.00	0.00	0.00	300
ङ	0.08	0.01	0.02	300
च	0.00	0.00	0.00	300
छ	0.00	0.00	0.00	300
ज	0.00	0.00	0.00	300
झ	0.03	0.03	0.03	300
ञ	0.09	0.92	0.17	300
ट	0.02	0.02	0.02	300
ठ	0.02	0.02	0.02	300
ड	0.00	0.00	0.00	300
ढ	0.10	0.96	0.18	300
ण	0.00	0.00	0.00	300
त	0.00	0.00	0.00	300
थ	0.00	0.00	0.00	300
द	0.00	0.00	0.00	300
ध	0.00	0.00	0.00	300
न	0.05	0.05	0.05	300
प	0.00	0.00	0.00	300
फ	0.00	0.00	0.00	300
ब	0.00	0.00	0.00	300
भ	0.02	0.02	0.02	300
म	0.05	0.09	0.07	300
य	0.00	0.00	0.00	300
र	0.00	0.00	0.00	300
ल	0.02	0.01	0.01	300

Class	Precision	Recall	F1-Score	Support
व	0.05	0.06	0.06	300
श	0.05	0.15	0.07	300
ष	0.00	0.00	0.00	300
स	0.09	0.10	0.09	300
ह	0.04	0.01	0.02	300
क्ष	0.04	0.05	0.04	300
त्र	0.00	0.00	0.00	300
ज्ञ	0.05	0.13	0.07	300

Table 8: Classification Report for Test Data ($n\_units = 5$)

Class	Precision	Recall	F1-Score	Support
क	0.49	0.69	0.57	300
ख	0.16	0.04	0.07	300
ग	0.36	0.31	0.34	300
घ	0.27	0.22	0.24	300
ङ	0.28	0.28	0.28	300
च	0.29	0.41	0.34	300
छ	0.37	0.46	0.41	300
ज	0.26	0.41	0.32	300
झ	0.47	0.71	0.57	300
ञ	0.45	0.49	0.47	300
ट	0.61	0.68	0.64	300
ठ	0.48	0.57	0.52	300
ड	0.33	0.30	0.31	300
ढ	0.40	0.71	0.51	300
ण	0.37	0.46	0.41	300
त	0.45	0.42	0.43	300
थ	0.26	0.30	0.28	300
द	0.21	0.11	0.14	300
ध	0.34	0.56	0.42	300
न	0.22	0.21	0.21	300
प	0.33	0.41	0.36	300
फ	0.63	0.70	0.66	300
ब	0.27	0.30	0.28	300
भ	0.39	0.52	0.45	300
म	0.27	0.08	0.13	300
य	0.16	0.06	0.08	300
र	0.42	0.38	0.40	300
ल	0.36	0.30	0.33	300
व	0.18	0.08	0.11	300
श	0.27	0.34	0.30	300

Class	Precision	Recall	F1-Score	Support
ष	0.33	0.34	0.33	300
स	0.28	0.03	0.05	300
ह	0.13	0.04	0.06	300
क्ष	0.29	0.25	0.27	300
त्र	0.19	0.11	0.14	300
ज्ञ	0.29	0.47	0.36	300

Table 9: Classification Report for Test Data ($n\_units = 10$)

Class	Precision	Recall	F1-Score	Support
क	0.66	0.67	0.67	300
ख	0.32	0.27	0.29	300
ग	0.53	0.65	0.59	300
घ	0.47	0.54	0.50	300
ङ	0.46	0.41	0.43	300
च	0.59	0.63	0.61	300
छ	0.53	0.52	0.52	300
ज	0.61	0.59	0.60	300
झ	0.72	0.73	0.73	300
ञ	0.66	0.74	0.69	300
ट	0.74	0.75	0.75	300
ठ	0.71	0.70	0.71	300
ड	0.59	0.56	0.57	300
ढ	0.63	0.68	0.65	300
ण	0.54	0.59	0.56	300
त	0.61	0.71	0.66	300
थ	0.46	0.46	0.46	300
द	0.58	0.49	0.53	300
ध	0.61	0.60	0.60	300
न	0.50	0.45	0.47	300
प	0.49	0.53	0.51	300
फ	0.73	0.74	0.74	300
ब	0.50	0.42	0.46	300
भ	0.54	0.60	0.57	300
म	0.38	0.25	0.30	300
य	0.36	0.25	0.29	300
र	0.70	0.70	0.70	300
ल	0.70	0.61	0.65	300
व	0.45	0.42	0.43	300
श	0.43	0.57	0.49	300
ष	0.52	0.69	0.59	300
स	0.44	0.36	0.40	300

Class	Precision	Recall	F1-Score	Support
ह	0.50	0.43	0.46	300
क्ष	0.42	0.45	0.43	300
त्र	0.51	0.51	0.51	300
ज्ञ	0.62	0.64	0.63	300

Table 10: Classification Report for Test Data ($n\_units = 50$)

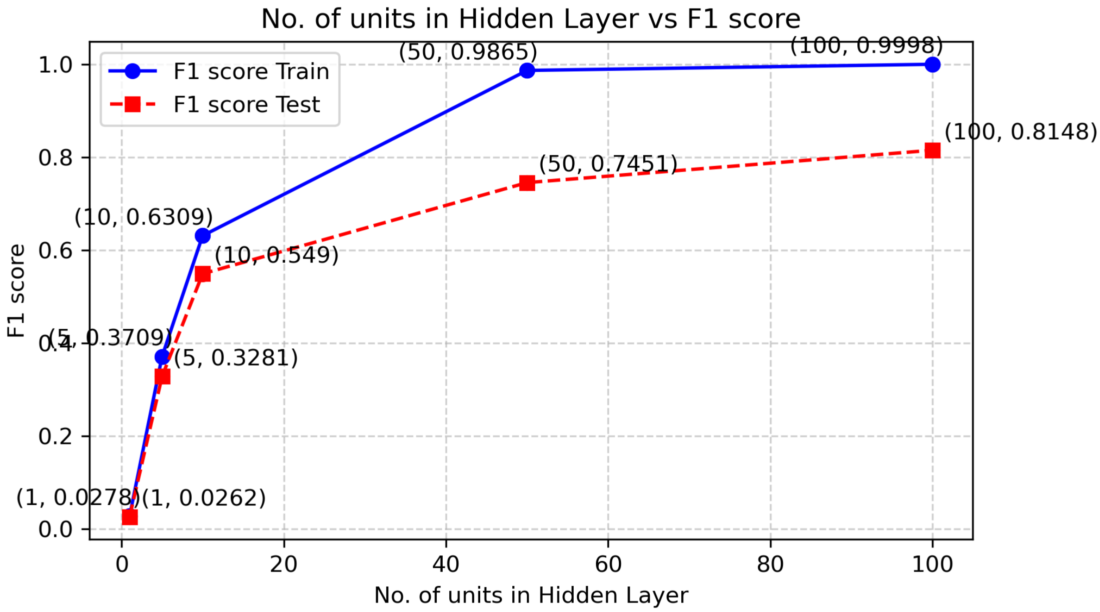
Class	Precision	Recall	F1-Score	Support
क	0.80	0.82	0.81	300
ख	0.70	0.67	0.68	300
ग	0.78	0.76	0.77	300
घ	0.67	0.69	0.68	300
ङ	0.69	0.70	0.70	300
च	0.80	0.78	0.79	300
छ	0.71	0.70	0.70	300
ज	0.77	0.81	0.79	300
झ	0.86	0.85	0.85	300
ञ	0.79	0.82	0.81	300
ट	0.91	0.87	0.89	300
ठ	0.85	0.82	0.83	300
ड	0.73	0.73	0.73	300
ढ	0.81	0.86	0.83	300
ण	0.78	0.82	0.80	300
त	0.81	0.85	0.83	300
थ	0.69	0.64	0.66	300
द	0.69	0.72	0.70	300
ध	0.72	0.72	0.72	300
न	0.65	0.69	0.67	300
प	0.69	0.71	0.70	300
फ	0.85	0.84	0.84	300
ब	0.69	0.67	0.68	300
भ	0.76	0.74	0.75	300
म	0.64	0.62	0.63	300
य	0.61	0.63	0.62	300
र	0.87	0.85	0.86	300
ल	0.88	0.78	0.83	300
व	0.66	0.67	0.66	300
श	0.68	0.74	0.71	300
ष	0.76	0.79	0.78	300
स	0.64	0.60	0.62	300
ह	0.68	0.70	0.69	300
क्ष	0.72	0.74	0.73	300

Class	Precision	Recall	F1-Score	Support
त्र	0.68	0.71	0.70	300
ज्ञ	0.81	0.72	0.76	300

Table 11: Classification Report for Test Data ($n\_units = 100$)

Class	Precision	Recall	F1-Score	Support
क	0.90	0.88	0.89	300
ख	0.73	0.74	0.73	300
ग	0.82	0.80	0.81	300
घ	0.73	0.78	0.75	300
ङ	0.81	0.77	0.79	300
च	0.83	0.86	0.84	300
छ	0.77	0.75	0.76	300
ज	0.83	0.87	0.85	300
झ	0.89	0.90	0.89	300
ञ	0.88	0.89	0.88	300
ट	0.92	0.92	0.92	300
ठ	0.91	0.91	0.91	300
ड	0.80	0.81	0.80	300
ढ	0.84	0.89	0.87	300
ण	0.84	0.87	0.86	300
त	0.86	0.92	0.89	300
थ	0.74	0.73	0.73	300
द	0.77	0.77	0.77	300
ध	0.74	0.74	0.74	300
न	0.79	0.76	0.77	300
प	0.81	0.81	0.81	300
फ	0.89	0.90	0.90	300
ब	0.71	0.73	0.72	300
भ	0.81	0.80	0.81	300
म	0.80	0.78	0.79	300
य	0.74	0.73	0.73	300
र	0.89	0.86	0.88	300
ल	0.89	0.84	0.87	300
व	0.76	0.70	0.73	300
श	0.77	0.80	0.79	300
ष	0.80	0.86	0.83	300
स	0.78	0.69	0.73	300
ह	0.80	0.78	0.79	300
क्ष	0.82	0.85	0.83	300
त्र	0.82	0.83	0.82	300
ज्ञ	0.86	0.82	0.84	300

The following plot shows the average f1-score vs number of hidden layer units.



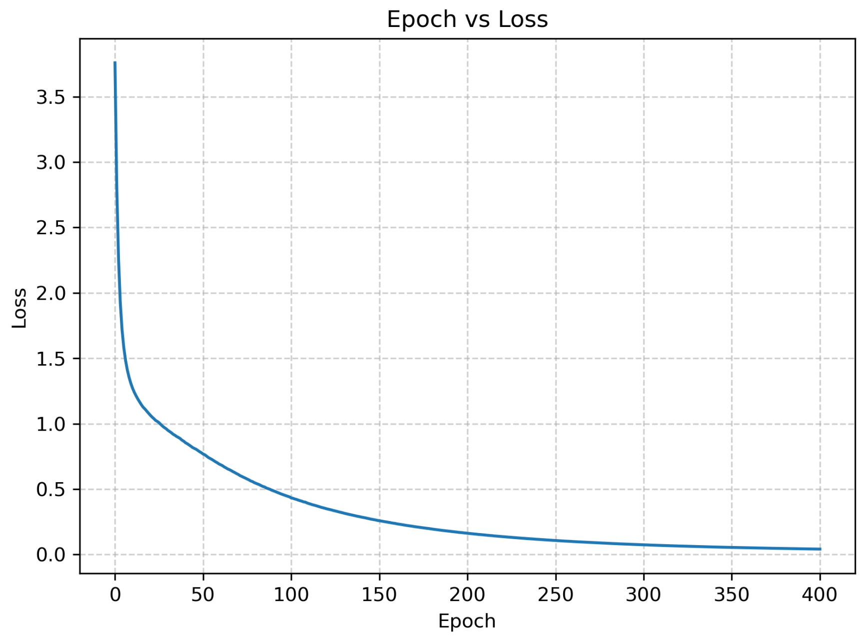
The model with $n\_units = 1$ performs very poorly. It gives close to baseline performance. For $n\_units = [5, 10]$ also it performs poorly on both train and test data indicating that a more complex model is needed.
For $n\_units = [50, 100]$ the model performs significantly better, the train f1-scores are close to 1 and test f1-scores are **0.745** and **0.815** respectively. The best single hidden layer model is the one with $n\_units = 100$ but it is overfitting, it is as if it has memorized the training data.
From the classification reports, we can say the letter ट is the most easily recognized by the model as it has the highest F1 score in the top 2 models.

(c) In this part we will vary the depth of the neural network and also the number of units in each layer. The architecture will be chosen as $\{\{512\}, \{512, 256\}, \{512, 256, 128\}, \{512, 256, 128, 64\}\}$. As initializing the model parameters as 0 causes the model to not learn so we use Xavier initialization that randomly initialized the parameters for each layer from

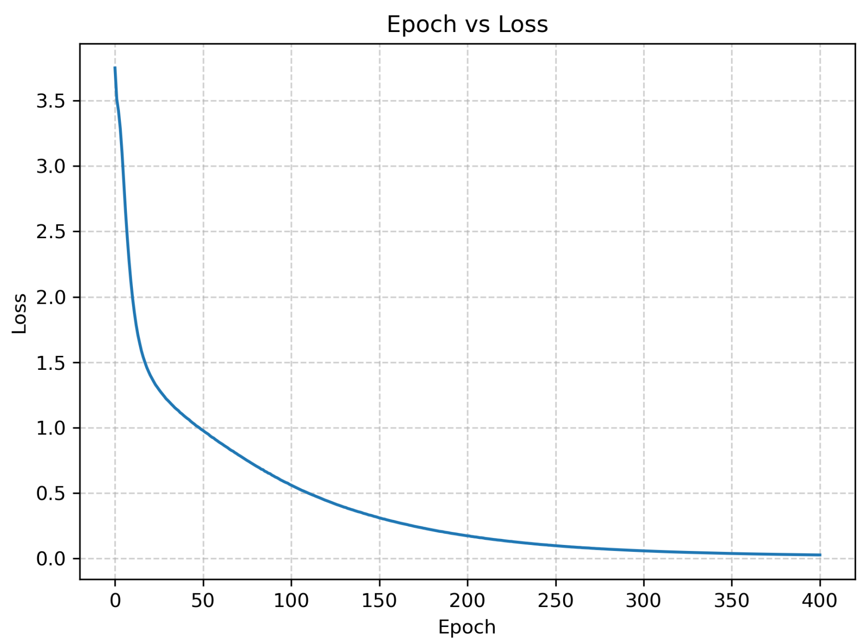
$$U(-\sqrt{6/(n_{in} + n_{out})}, \sqrt{6/(n_{in} + n_{out})})$$

.

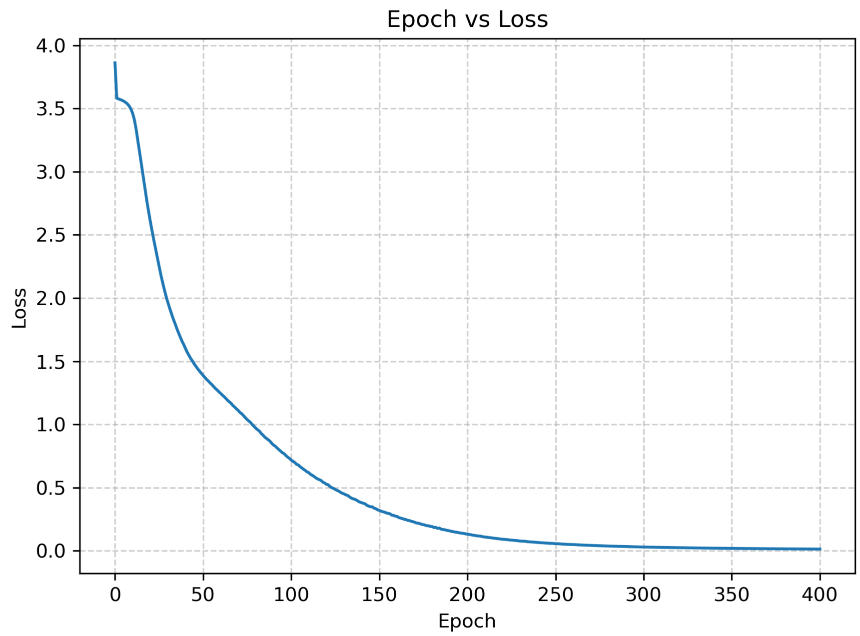
The following learning curves were obtained.



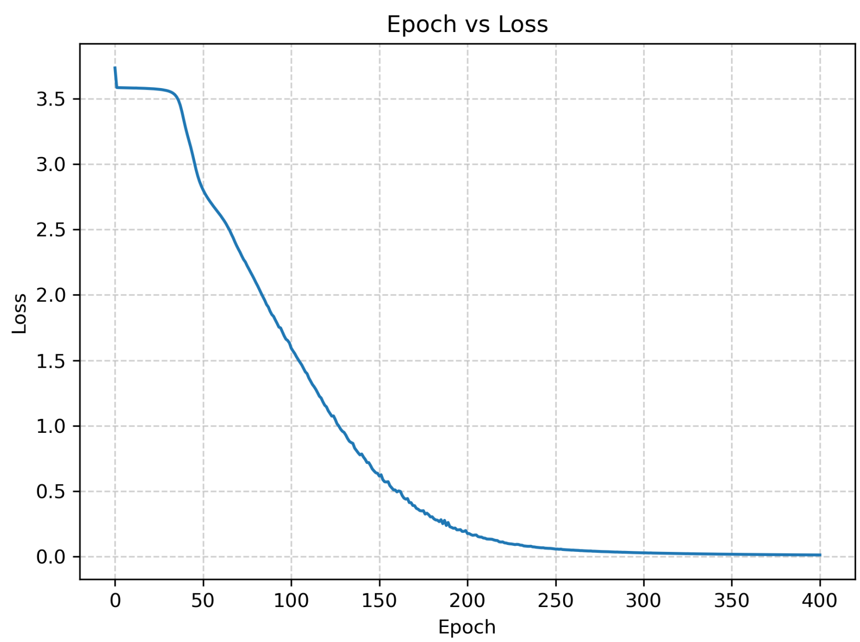
depth $= 1$



depth $= 2$



depth = 3



depth = 4

The following *Classification Reports* were obtained for training data.

Table 12: Classification Report for Training Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 13: Classification Report for Training Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 14: Classification Report for Training Data ($depth = 3$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 15: Classification Report for Training Data ($depth = 4$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

The following classification reports were obtained for Test data

Table 16: Classification Report for Test Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	0.94	0.96	0.95	300
ख	0.88	0.86	0.87	300
ग	0.89	0.87	0.88	300
घ	0.83	0.82	0.83	300
ङ	0.87	0.85	0.86	300
च	0.89	0.90	0.90	300
छ	0.88	0.85	0.86	300
ज	0.88	0.91	0.90	300
झ	0.93	0.93	0.93	300
ञ	0.93	0.91	0.92	300
ट	0.95	0.96	0.96	300
ठ	0.93	0.93	0.93	300
ड	0.88	0.90	0.89	300
ढ	0.88	0.92	0.90	300
ण	0.90	0.94	0.92	300
त	0.92	0.96	0.94	300
थ	0.86	0.81	0.83	300
द	0.88	0.87	0.87	300
ध	0.84	0.84	0.84	300
न	0.84	0.85	0.84	300
प	0.84	0.89	0.87	300
फ	0.94	0.94	0.94	300
ब	0.82	0.84	0.83	300
भ	0.86	0.87	0.87	300
म	0.87	0.85	0.86	300
य	0.88	0.84	0.86	300
र	0.96	0.94	0.95	300
ल	0.93	0.94	0.93	300
व	0.87	0.88	0.87	300
श	0.89	0.89	0.89	300
ष	0.85	0.92	0.89	300
स	0.88	0.81	0.84	300
ह	0.90	0.87	0.88	300
क्ष	0.89	0.91	0.90	300
त्र	0.91	0.88	0.90	300
ज्ञ	0.89	0.88	0.88	300

Table 17: Classification Report for Test Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	0.94	0.95	0.95	300

Class	Precision	Recall	F1-Score	Support
ख	0.89	0.86	0.88	300
ग	0.92	0.89	0.91	300
घ	0.82	0.83	0.82	300
ङ	0.88	0.87	0.87	300
च	0.88	0.91	0.90	300
छ	0.86	0.86	0.86	300
ज	0.91	0.92	0.91	300
झ	0.96	0.95	0.95	300
ञ	0.93	0.95	0.94	300
ट	0.96	0.95	0.96	300
ठ	0.95	0.95	0.95	300
ड	0.88	0.91	0.89	300
ढ	0.90	0.94	0.92	300
ण	0.92	0.95	0.93	300
त	0.94	0.96	0.95	300
थ	0.87	0.79	0.83	300
द	0.90	0.86	0.88	300
ध	0.83	0.84	0.83	300
न	0.87	0.87	0.87	300
प	0.85	0.89	0.87	300
फ	0.94	0.94	0.94	300
ब	0.83	0.82	0.82	300
भ	0.89	0.87	0.88	300
म	0.87	0.87	0.87	300
य	0.87	0.86	0.86	300
र	0.95	0.95	0.95	300
ल	0.90	0.93	0.92	300
व	0.85	0.88	0.87	300
श	0.94	0.91	0.92	300
ष	0.86	0.90	0.88	300
स	0.91	0.86	0.88	300
ह	0.88	0.86	0.87	300
क्ष	0.89	0.91	0.90	300
त्र	0.91	0.89	0.90	300
ज्ञ	0.93	0.89	0.91	300

Table 18: Classification Report for Test Data ($depth = 3$)

Class	Precision	Recall	F1-Score	Support
क	0.92	0.93	0.93	300
ख	0.90	0.85	0.87	300
ग	0.91	0.88	0.89	300
घ	0.81	0.86	0.84	300

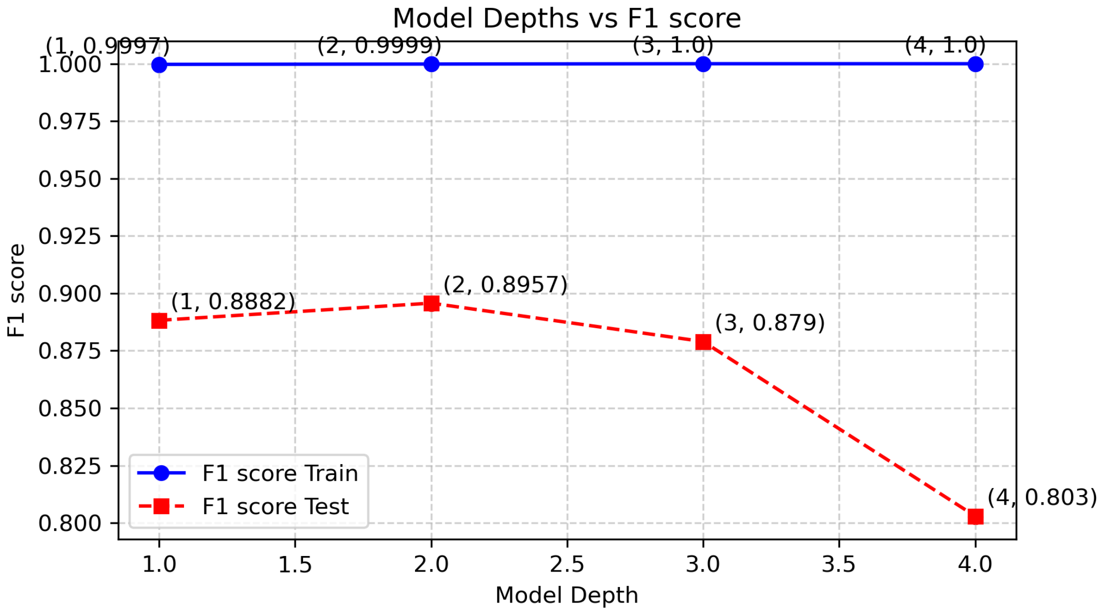
Class	Precision	Recall	F1-Score	Support
ङ	0.88	0.84	0.86	300
च	0.87	0.89	0.88	300
छ	0.88	0.88	0.88	300
ज	0.91	0.90	0.90	300
झ	0.94	0.93	0.93	300
ञ	0.90	0.91	0.91	300
ट	0.95	0.95	0.95	300
ठ	0.92	0.95	0.94	300
ड	0.86	0.88	0.87	300
ढ	0.89	0.92	0.90	300
ण	0.85	0.91	0.88	300
त	0.94	0.95	0.94	300
थ	0.83	0.81	0.82	300
द	0.88	0.84	0.86	300
ध	0.81	0.82	0.82	300
न	0.88	0.85	0.87	300
प	0.83	0.87	0.85	300
फ	0.92	0.93	0.93	300
ब	0.84	0.81	0.82	300
भ	0.85	0.84	0.85	300
म	0.85	0.84	0.84	300
य	0.83	0.82	0.83	300
र	0.94	0.93	0.93	300
ल	0.89	0.93	0.91	300
व	0.86	0.87	0.86	300
श	0.88	0.90	0.89	300
ष	0.87	0.88	0.87	300
स	0.85	0.84	0.85	300
ह	0.86	0.82	0.84	300
क्ष	0.87	0.88	0.87	300
त्र	0.90	0.89	0.90	300
ज्ञ	0.90	0.88	0.89	300

Table 19: Classification Report for Test Data ($depth = 4$)

Class	Precision	Recall	F1-Score	Support
क	0.90	0.89	0.89	300
ख	0.78	0.67	0.72	300
ग	0.80	0.78	0.79	300
घ	0.72	0.75	0.73	300
ङ	0.77	0.73	0.75	300
च	0.89	0.80	0.84	300
छ	0.80	0.77	0.78	300

Class	Precision	Recall	F1-Score	Support
ज	0.81	0.86	0.83	300
झ	0.91	0.92	0.92	300
ञ	0.82	0.88	0.85	300
ट	0.89	0.93	0.91	300
ठ	0.87	0.89	0.88	300
ड	0.83	0.78	0.80	300
ढ	0.83	0.88	0.85	300
ण	0.80	0.78	0.79	300
त	0.84	0.91	0.87	300
थ	0.78	0.79	0.79	300
द	0.79	0.80	0.80	300
ध	0.81	0.80	0.81	300
न	0.80	0.75	0.78	300
प	0.72	0.81	0.77	300
फ	0.83	0.90	0.87	300
ब	0.75	0.63	0.68	300
भ	0.81	0.82	0.81	300
म	0.74	0.77	0.76	300
य	0.63	0.62	0.63	300
र	0.84	0.92	0.88	300
ल	0.81	0.85	0.83	300
व	0.79	0.69	0.74	300
श	0.82	0.83	0.82	300
ष	0.76	0.83	0.80	300
स	0.75	0.73	0.74	300
ह	0.77	0.75	0.76	300
क्ष	0.82	0.81	0.82	300
त्र	0.80	0.80	0.80	300
ज्ञ	0.87	0.81	0.84	300

The following plot shows the F1-score vs the model depth

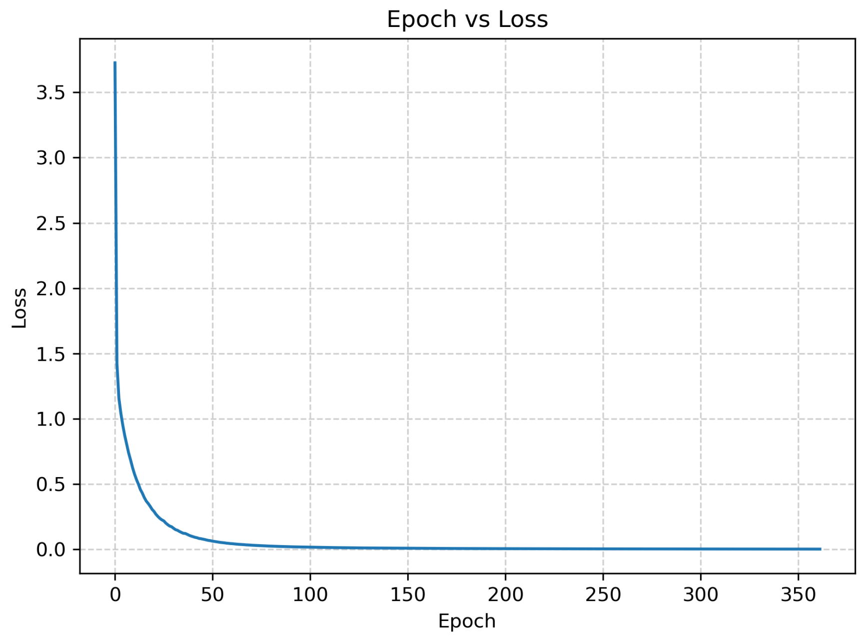


All the models have almost memorized the training data. The model with $depth = 2$ performs the best on the test data with close to 0.9 f1-score. This is an improvement form the models in the previous part. Here also, the letter ट is the most easily recognized letter. The letter ख is not that easy to recognize for the model because it is often written as रव in the dataset.

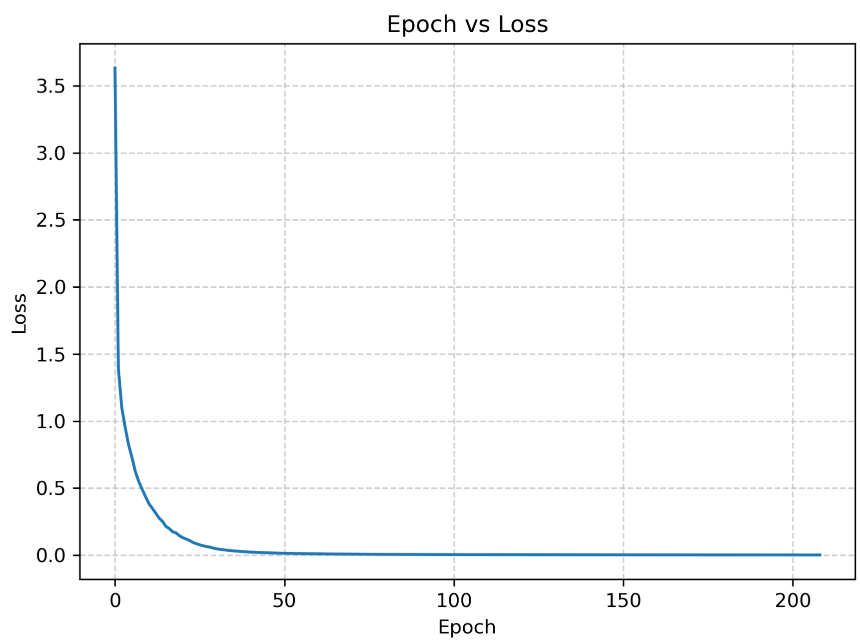
Overall having a multilayer architecture is better. Too many layers lead to loss in generalization.

(d) In this everything is done same as part (c) only the activations of the hidden layers are changed to ReLU from sigmoid activation.

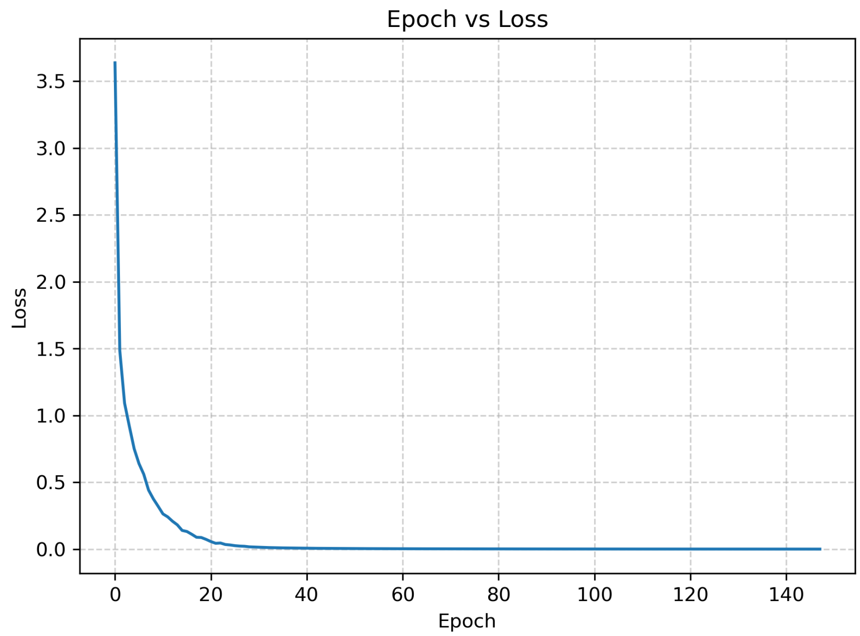
The following learning curves were obtained.



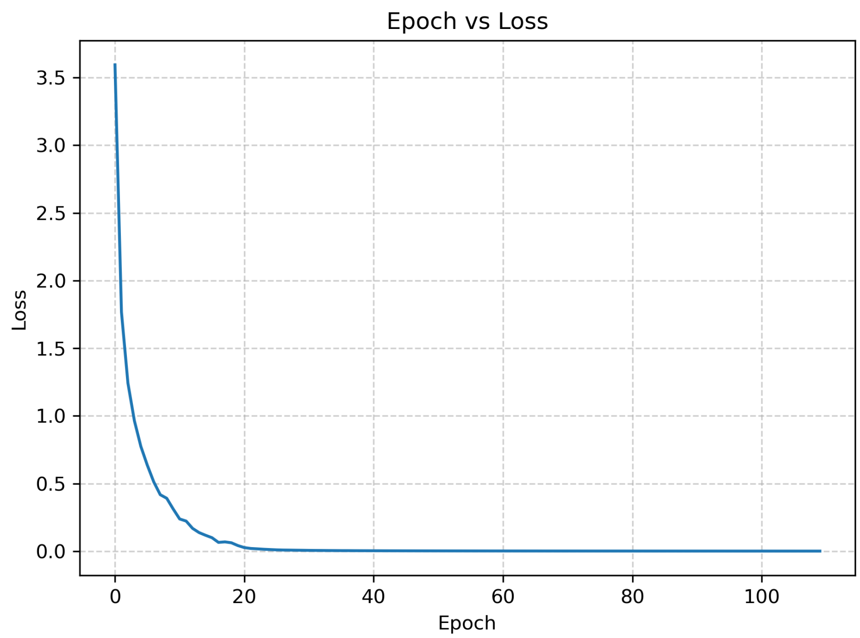
depth = 1



depth = 2



depth = 3



depth = 4

The models converge faster than part (c) when sigmoid activation was being used. The following *Classification Reports* were obtained for training data.

Table 20: Classification Report for Training Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 21: Classification Report for Training Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 22: Classification Report for Training Data ($depth = 3$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 23: Classification Report for Training Data ($depth = 4$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

The following classification reports were obtained for Test data

Table 24: Classification Report for Test Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	0.95	0.96	0.96	300
ख	0.90	0.88	0.89	300
ग	0.93	0.90	0.91	300
घ	0.85	0.84	0.84	300
ङ	0.89	0.87	0.88	300
च	0.90	0.91	0.90	300
छ	0.90	0.87	0.88	300
ज	0.89	0.91	0.90	300
झ	0.95	0.94	0.94	300
ञ	0.95	0.94	0.95	300
ट	0.95	0.94	0.94	300
ठ	0.91	0.94	0.93	300
ड	0.87	0.89	0.88	300
ढ	0.93	0.92	0.93	300
ण	0.90	0.93	0.92	300
त	0.91	0.95	0.93	300
थ	0.83	0.80	0.82	300
द	0.87	0.87	0.87	300
ध	0.85	0.84	0.84	300
न	0.87	0.87	0.87	300
प	0.84	0.89	0.87	300
फ	0.92	0.94	0.93	300
ब	0.84	0.82	0.83	300
भ	0.89	0.88	0.89	300
म	0.87	0.89	0.88	300
य	0.86	0.85	0.85	300
र	0.96	0.94	0.95	300
ल	0.95	0.93	0.94	300
व	0.85	0.87	0.86	300
श	0.88	0.93	0.90	300
ष	0.90	0.91	0.91	300
स	0.90	0.83	0.86	300
ह	0.90	0.89	0.89	300
क्ष	0.91	0.93	0.92	300
त्र	0.92	0.90	0.91	300
ज्ञ	0.91	0.91	0.91	300

Table 25: Classification Report for Test Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	0.93	0.95	0.94	300

Class	Precision	Recall	F1-Score	Support
ख	0.87	0.88	0.88	300
ग	0.90	0.90	0.90	300
घ	0.84	0.84	0.84	300
ङ	0.91	0.85	0.88	300
च	0.90	0.91	0.90	300
छ	0.89	0.89	0.89	300
ज	0.90	0.89	0.90	300
झ	0.96	0.95	0.95	300
ञ	0.93	0.94	0.94	300
ट	0.96	0.95	0.95	300
ठ	0.92	0.95	0.93	300
ड	0.88	0.90	0.89	300
ढ	0.90	0.94	0.92	300
ण	0.90	0.93	0.91	300
त	0.94	0.94	0.94	300
थ	0.84	0.79	0.82	300
द	0.90	0.88	0.89	300
ध	0.84	0.85	0.85	300
न	0.82	0.86	0.84	300
प	0.86	0.90	0.88	300
फ	0.94	0.96	0.95	300
ब	0.87	0.83	0.85	300
भ	0.92	0.87	0.90	300
म	0.88	0.89	0.88	300
य	0.86	0.85	0.85	300
र	0.96	0.94	0.95	300
ल	0.92	0.93	0.93	300
व	0.86	0.87	0.87	300
श	0.88	0.91	0.90	300
ष	0.89	0.90	0.90	300
स	0.91	0.82	0.86	300
ह	0.91	0.87	0.89	300
क्ष	0.86	0.92	0.89	300
त्र	0.92	0.91	0.91	300
ज्ञ	0.93	0.91	0.92	300

Table 26: Classification Report for Test Data ($depth = 3$)

Class	Precision	Recall	F1-Score	Support
क	0.94	0.94	0.94	300
ख	0.91	0.86	0.89	300
ग	0.90	0.89	0.90	300
घ	0.84	0.85	0.84	300

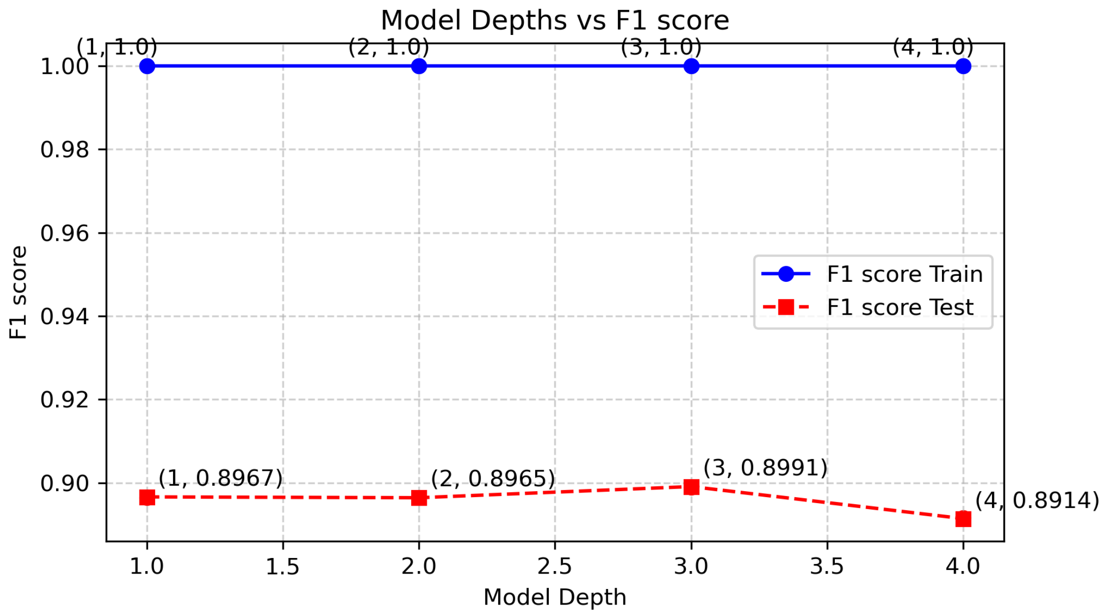
Class	Precision	Recall	F1-Score	Support
ङ	0.87	0.85	0.86	300
च	0.89	0.92	0.91	300
छ	0.89	0.86	0.87	300
ज	0.91	0.92	0.92	300
झ	0.97	0.95	0.96	300
ञ	0.94	0.95	0.94	300
ट	0.96	0.96	0.96	300
ठ	0.93	0.95	0.94	300
ड	0.89	0.91	0.90	300
ढ	0.89	0.94	0.92	300
ण	0.91	0.94	0.92	300
त	0.94	0.97	0.95	300
थ	0.86	0.84	0.85	300
द	0.89	0.87	0.88	300
ध	0.85	0.86	0.86	300
न	0.83	0.86	0.84	300
प	0.88	0.88	0.88	300
फ	0.94	0.94	0.94	300
ब	0.86	0.81	0.84	300
भ	0.90	0.89	0.89	300
म	0.85	0.91	0.88	300
य	0.84	0.83	0.84	300
र	0.96	0.94	0.95	300
ल	0.93	0.94	0.94	300
व	0.88	0.87	0.87	300
श	0.89	0.91	0.90	300
ष	0.88	0.91	0.90	300
स	0.90	0.84	0.87	300
ह	0.92	0.89	0.91	300
क्ष	0.88	0.92	0.90	300
त्र	0.93	0.91	0.92	300
ज्ञ	0.92	0.90	0.91	300

Table 27: Classification Report for Test Data ($depth = 4$)

Class	Precision	Recall	F1-Score	Support
क	0.93	0.92	0.92	300
ख	0.88	0.87	0.88	300
ग	0.90	0.91	0.91	300
घ	0.84	0.86	0.85	300
ङ	0.88	0.83	0.85	300
च	0.88	0.92	0.90	300
छ	0.90	0.87	0.88	300

Class	Precision	Recall	F1-Score	Support
ज	0.93	0.89	0.91	300
झ	0.94	0.93	0.93	300
ञ	0.94	0.93	0.93	300
ट	0.96	0.96	0.96	300
ठ	0.93	0.93	0.93	300
ड	0.87	0.90	0.89	300
ढ	0.91	0.93	0.92	300
ण	0.92	0.95	0.93	300
त	0.92	0.93	0.93	300
थ	0.88	0.80	0.84	300
द	0.91	0.87	0.89	300
ध	0.84	0.87	0.85	300
न	0.85	0.87	0.86	300
प	0.88	0.88	0.88	300
फ	0.94	0.94	0.94	300
ब	0.82	0.82	0.82	300
भ	0.88	0.87	0.88	300
म	0.83	0.86	0.84	300
य	0.83	0.86	0.84	300
र	0.95	0.94	0.94	300
ल	0.92	0.91	0.91	300
व	0.86	0.87	0.86	300
श	0.90	0.88	0.89	300
ष	0.86	0.91	0.88	300
स	0.86	0.83	0.84	300
ह	0.88	0.88	0.88	300
क्ष	0.88	0.90	0.89	300
त्र	0.89	0.91	0.90	300
ज्ञ	0.91	0.90	0.91	300

The following plot shows the F1-score vs the model depth

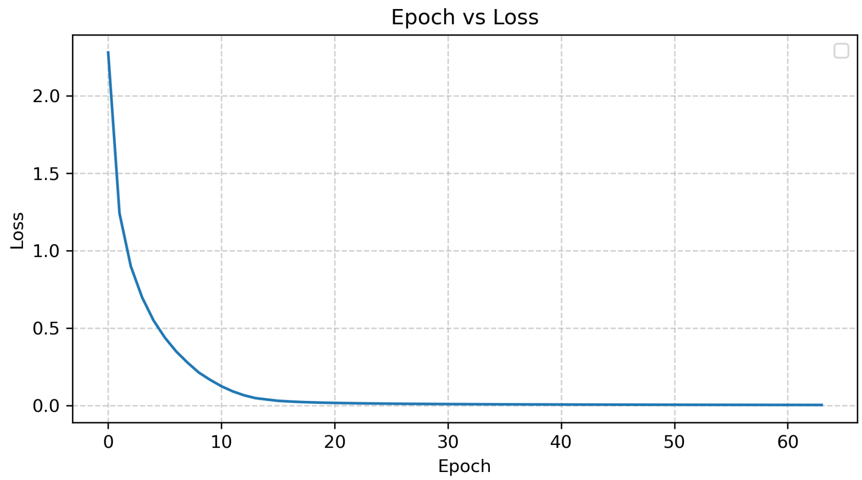


Here the best model is the one with $depth = 3$ with a training accuracy f1-score of 0.8991 which is slightly higher than the one in part (c)2.3. All the models have similar test accuracies here whereas in the previous case all had significant variation in test accuracies across the various depths and architectures. As the test dataset has significant size even slight improvements indicate a better model, so using ReLU activation for all the hidden layers gives better performance.

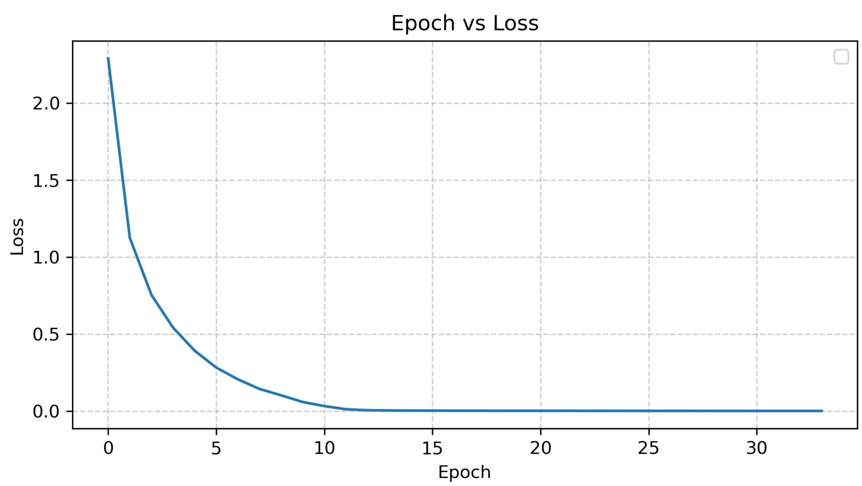
(e) In this part, sci-kit learn's MLPClassifier is used with following parameters

- **Hidden layer sizes:** to be varied according to part (c)
- **Activation:** ReLU
- **Solver:** SGD
- **Alpha:** 0
- **Batch size:** 32
- **Learning rate:** Constant
- **max_iter:** 400

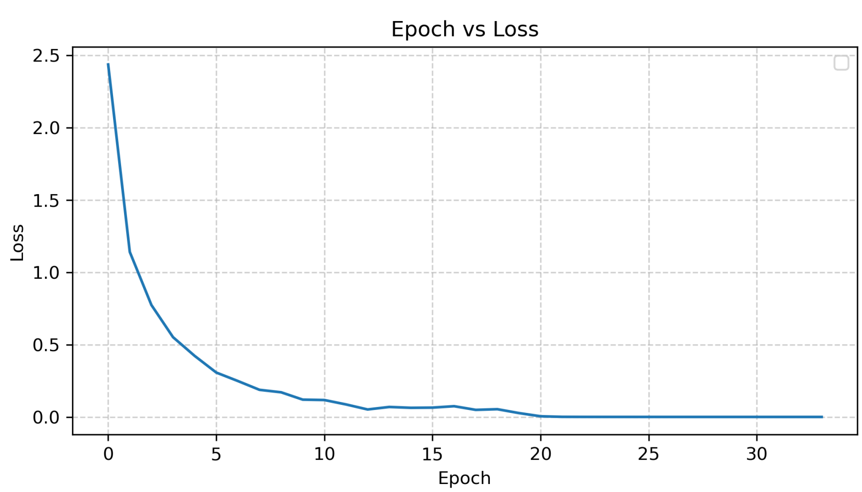
The following learning curves were obtained.



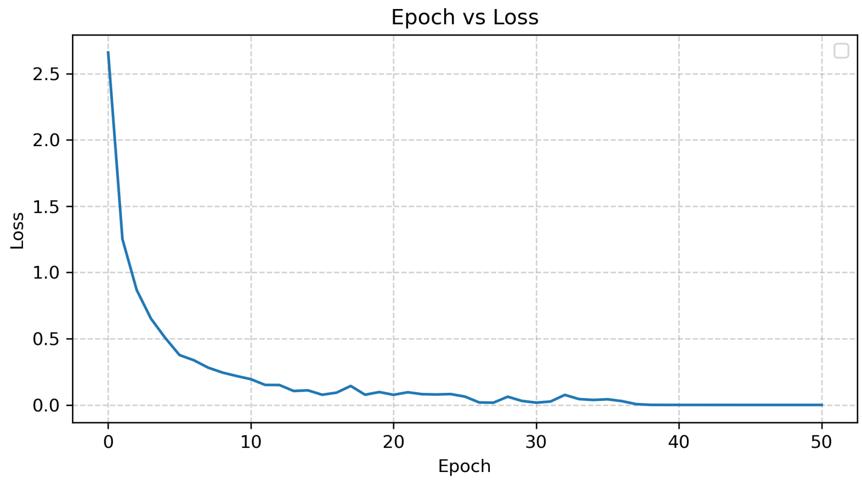
depth = 1



depth = 2



depth = 3



depth = 4

The models converge faster than part (d). This probably happens because of different initialization strategy used in sklearn
The following *Classification Reports* were obtained for training data.

Table 28: Classification Report for Training Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600

Class	Precision	Recall	F1-Score	Support
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 29: Classification Report for Training Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600

Class	Precision	Recall	F1-Score	Support
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 30: Classification Report for Training Data ($depth = 3$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600

Class	Precision	Recall	F1-Score	Support
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

Table 31: Classification Report for Training Data ($depth = 4$)

Class	Precision	Recall	F1-Score	Support
क	1.00	1.00	1.00	600
ख	1.00	1.00	1.00	600
ग	1.00	1.00	1.00	600
घ	1.00	1.00	1.00	600
ङ	1.00	1.00	1.00	600
च	1.00	1.00	1.00	600
छ	1.00	1.00	1.00	600
ज	1.00	1.00	1.00	600
झ	1.00	1.00	1.00	600
ञ	1.00	1.00	1.00	600
ट	1.00	1.00	1.00	600
ठ	1.00	1.00	1.00	600
ड	1.00	1.00	1.00	600
ढ	1.00	1.00	1.00	600
ण	1.00	1.00	1.00	600
त	1.00	1.00	1.00	600
थ	1.00	1.00	1.00	600
द	1.00	1.00	1.00	600
ध	1.00	1.00	1.00	600
न	1.00	1.00	1.00	600
प	1.00	1.00	1.00	600
फ	1.00	1.00	1.00	600
ब	1.00	1.00	1.00	600
भ	1.00	1.00	1.00	600
म	1.00	1.00	1.00	600
य	1.00	1.00	1.00	600
र	1.00	1.00	1.00	600

Class	Precision	Recall	F1-Score	Support
ल	1.00	1.00	1.00	600
व	1.00	1.00	1.00	600
श	1.00	1.00	1.00	600
ष	1.00	1.00	1.00	600
स	1.00	1.00	1.00	600
ह	1.00	1.00	1.00	600
क्ष	1.00	1.00	1.00	600
त्र	1.00	1.00	1.00	600
ज्ञ	1.00	1.00	1.00	600

The following classification reports were obtained for Test data

Table 32: Classification Report for Test Data ($depth = 1$)

Class	Precision	Recall	F1-Score	Support
क	0.97	0.93	0.95	300
ख	0.97	0.84	0.90	300
ग	0.96	0.89	0.92	300
घ	0.87	0.76	0.81	300
ङ	0.94	0.79	0.86	300
च	0.97	0.87	0.92	300
छ	0.94	0.81	0.87	300
ज	0.92	0.87	0.89	300
झ	0.97	0.92	0.95	300
ञ	0.96	0.88	0.92	300
ट	0.98	0.95	0.96	300
ठ	0.97	0.91	0.94	300
ड	0.94	0.87	0.90	300
ढ	0.93	0.92	0.92	300
ण	0.93	0.90	0.92	300
त	0.96	0.93	0.94	300
थ	0.90	0.78	0.83	300
द	0.90	0.82	0.86	300
ध	0.86	0.79	0.82	300
न	0.90	0.81	0.86	300
प	0.92	0.83	0.87	300
फ	0.96	0.90	0.93	300
ब	0.92	0.77	0.84	300
भ	0.92	0.84	0.88	300
म	0.89	0.80	0.84	300
य	0.89	0.74	0.81	300
र	0.98	0.91	0.94	300
ल	0.97	0.89	0.93	300
व	0.91	0.81	0.86	300

Class	Precision	Recall	F1-Score	Support
श	0.94	0.86	0.90	300
ष	0.93	0.87	0.90	300
स	0.92	0.71	0.80	300
ह	0.96	0.83	0.89	300
क्ष	0.92	0.88	0.90	300
त्र	0.96	0.88	0.92	300
ज्ञ	0.95	0.84	0.89	300

Table 33: Classification Report for Test Data ($depth = 2$)

Class	Precision	Recall	F1-Score	Support
क	0.97	0.93	0.95	300
ख	0.97	0.87	0.92	300
ग	0.97	0.91	0.94	300
घ	0.89	0.79	0.84	300
ङ	0.92	0.82	0.87	300
च	0.95	0.88	0.92	300
छ	0.94	0.84	0.89	300
ज	0.95	0.90	0.93	300
झ	0.98	0.95	0.96	300
ञ	0.96	0.90	0.93	300
ट	0.98	0.96	0.97	300
ठ	0.98	0.92	0.95	300
ड	0.91	0.88	0.90	300
ढ	0.94	0.94	0.94	300
ण	0.95	0.93	0.94	300
त	0.95	0.94	0.95	300
थ	0.88	0.79	0.83	300
द	0.93	0.82	0.87	300
ध	0.90	0.83	0.86	300
न	0.93	0.88	0.90	300
प	0.93	0.89	0.91	300
फ	0.97	0.92	0.95	300
ब	0.90	0.81	0.85	300
भ	0.94	0.84	0.89	300
म	0.93	0.88	0.90	300
य	0.91	0.80	0.85	300
र	0.98	0.94	0.96	300
ल	0.96	0.92	0.94	300
व	0.92	0.86	0.89	300
श	0.95	0.87	0.91	300
ष	0.93	0.88	0.91	300
स	0.96	0.76	0.85	300

Class	Precision	Recall	F1-Score	Support
ह	0.94	0.86	0.90	300
क्ष	0.94	0.88	0.91	300
त्र	0.96	0.90	0.93	300
ज्ञ	0.95	0.88	0.91	300

Table 34: Classification Report for Test Data ($depth = 3$)

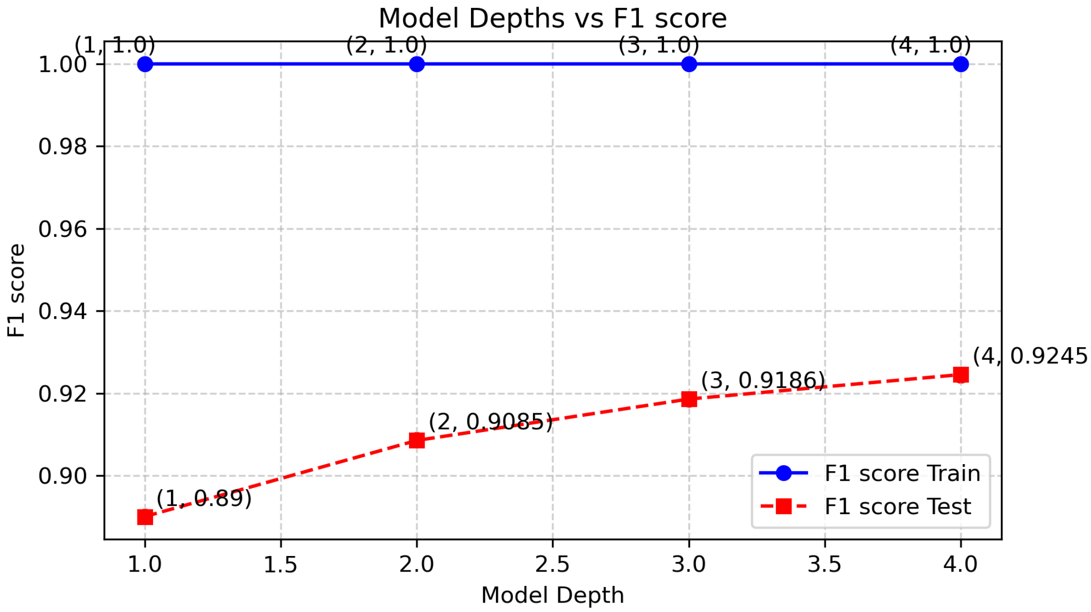
Class	Precision	Recall	F1-Score	Support
क	0.98	0.94	0.96	300
ख	0.98	0.87	0.92	300
ग	0.96	0.92	0.94	300
घ	0.88	0.82	0.85	300
ङ	0.93	0.83	0.87	300
च	0.96	0.91	0.93	300
छ	0.94	0.89	0.92	300
ज	0.94	0.91	0.92	300
झ	0.98	0.94	0.96	300
ञ	0.97	0.93	0.95	300
ट	0.99	0.96	0.98	300
ठ	0.96	0.95	0.95	300
ड	0.93	0.88	0.90	300
ढ	0.94	0.96	0.95	300
ण	0.95	0.94	0.94	300
त	0.96	0.96	0.96	300
थ	0.91	0.83	0.87	300
द	0.93	0.85	0.89	300
ध	0.94	0.87	0.90	300
न	0.93	0.88	0.91	300
प	0.91	0.86	0.89	300
फ	0.99	0.95	0.97	300
ब	0.94	0.83	0.88	300
भ	0.92	0.85	0.89	300
म	0.90	0.89	0.89	300
य	0.93	0.84	0.88	300
र	0.99	0.94	0.97	300
ल	0.97	0.92	0.94	300
व	0.93	0.85	0.89	300
श	0.97	0.89	0.93	300
ष	0.94	0.89	0.91	300
स	0.94	0.82	0.87	300
ह	0.96	0.86	0.91	300
क्ष	0.95	0.90	0.93	300
त्र	0.95	0.91	0.93	300

Class	Precision	Recall	F1-Score	Support
ज्ञ	0.95	0.90	0.92	300

Table 35: Classification Report for Test Data ($depth = 4$)

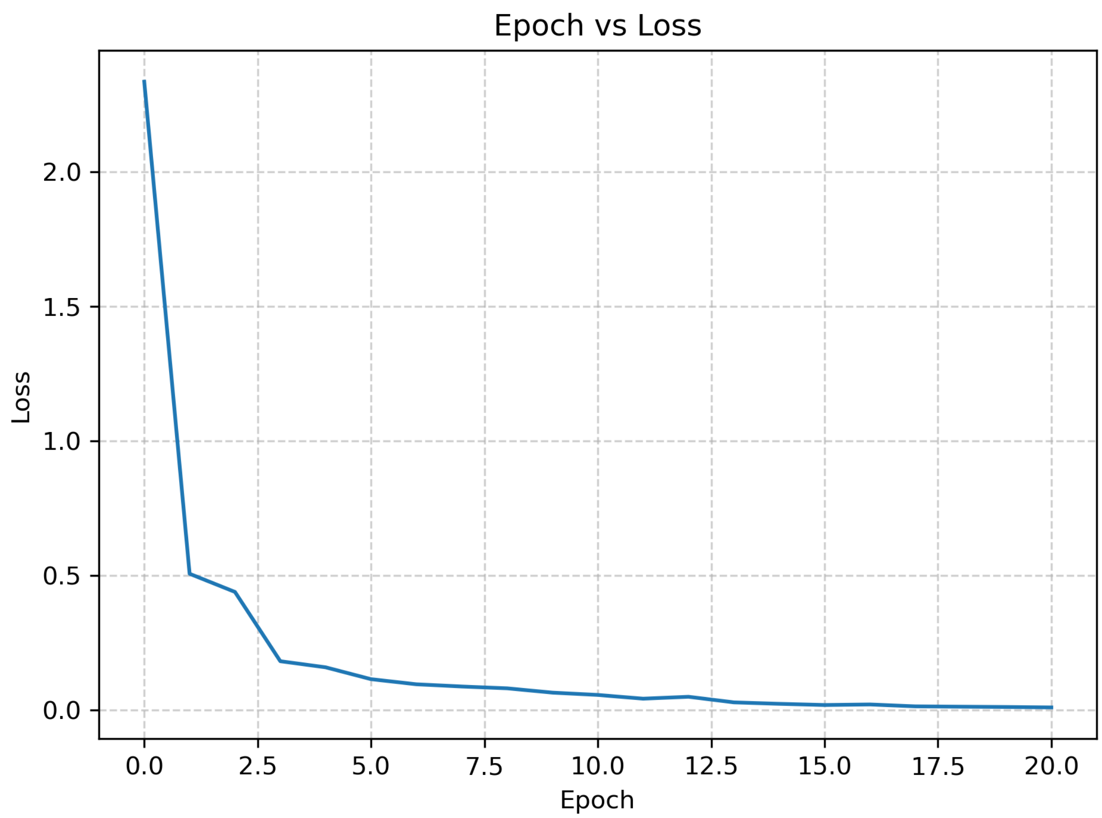
Class	Precision	Recall	F1-Score	Support
क	0.98	0.94	0.96	300
ख	0.96	0.89	0.92	300
ग	0.97	0.92	0.95	300
घ	0.87	0.87	0.87	300
ङ	0.93	0.86	0.89	300
च	0.95	0.91	0.93	300
छ	0.94	0.89	0.91	300
ज	0.94	0.93	0.93	300
झ	0.98	0.96	0.97	300
ञ	0.95	0.95	0.95	300
ट	0.98	0.96	0.97	300
ठ	0.97	0.95	0.96	300
ड	0.94	0.92	0.93	300
ढ	0.93	0.96	0.94	300
ण	0.93	0.96	0.95	300
त	0.96	0.96	0.96	300
थ	0.87	0.86	0.87	300
द	0.93	0.87	0.90	300
ध	0.91	0.87	0.89	300
न	0.92	0.92	0.92	300
प	0.89	0.92	0.90	300
फ	0.98	0.96	0.97	300
ब	0.90	0.87	0.88	300
भ	0.92	0.90	0.91	300
म	0.89	0.87	0.88	300
य	0.90	0.86	0.88	300
र	0.99	0.95	0.97	300
ल	0.96	0.94	0.95	300
व	0.90	0.90	0.90	300
श	0.96	0.92	0.94	300
ष	0.94	0.92	0.93	300
स	0.94	0.84	0.89	300
ह	0.97	0.91	0.94	300
क्ष	0.92	0.92	0.92	300
त्र	0.95	0.91	0.93	300
ज्ञ	0.95	0.92	0.94	300

The following plot shows the F1-score vs the model depth

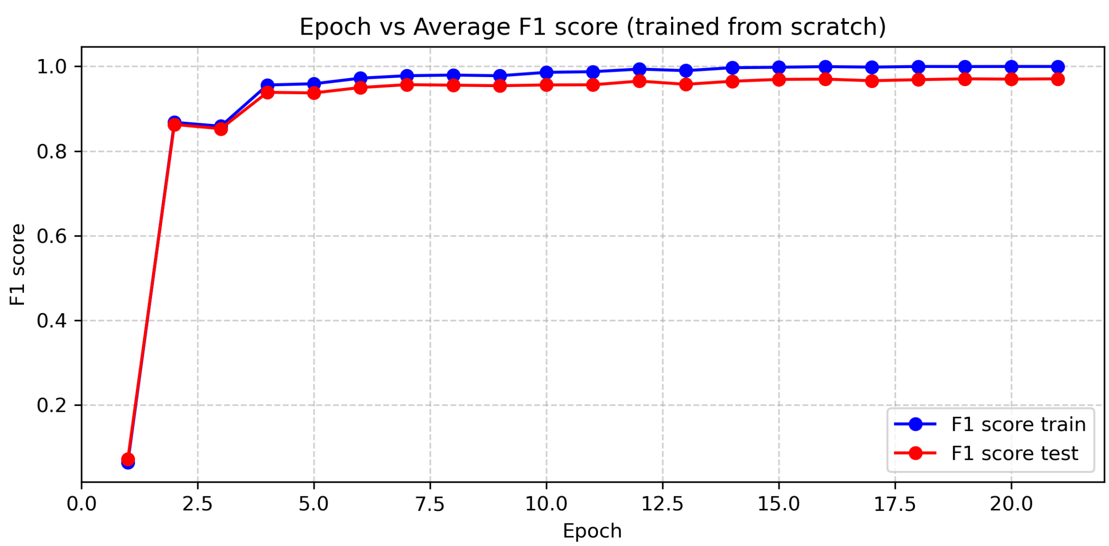


Here, the model with $depth = 4$ performs best with test f1-score as 0.9245. This is an improvement from the performance seen in part (d). This could be because there, the model has trained for too long leading to unnecessary overfitting.

(f)(i) In this part, a new dataset is taken containing digits of Hindi alphabet. The 4 layer architecture is used similar to those in above parts. 20 epochs were ran for the training. The following learning curve was obtained.

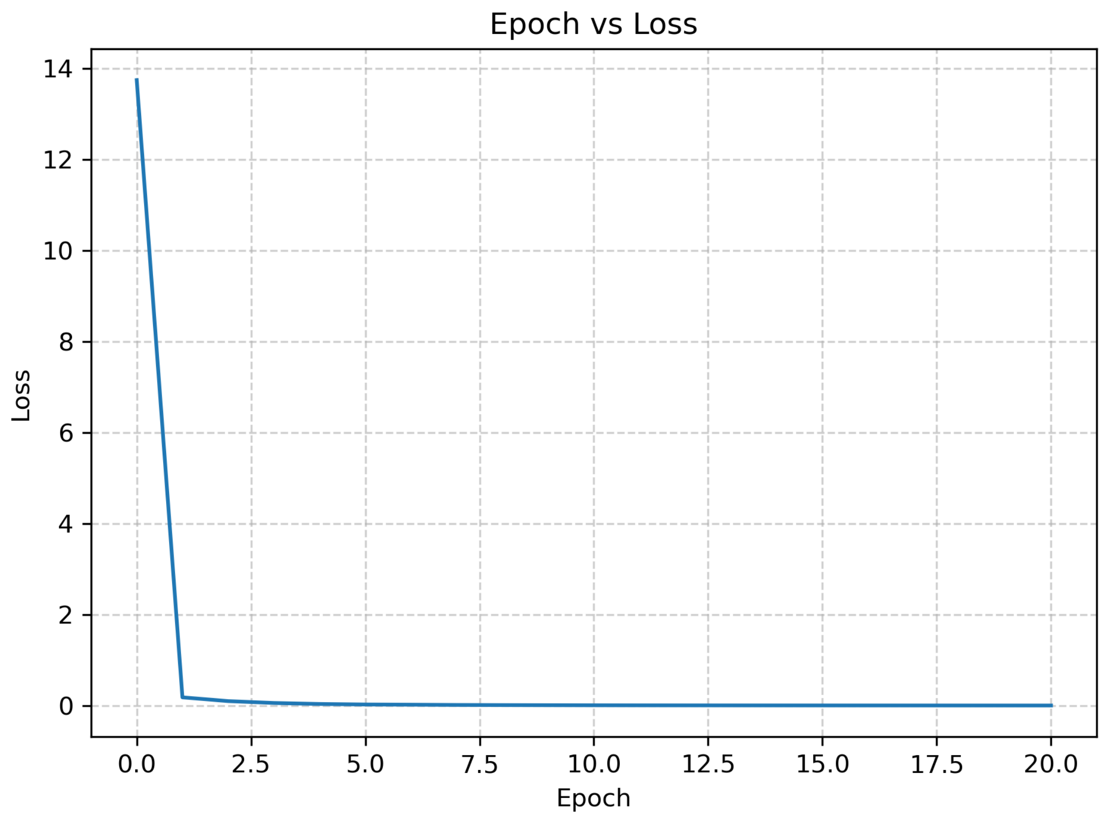


The following curve shows how the F1-score changes through the epochs

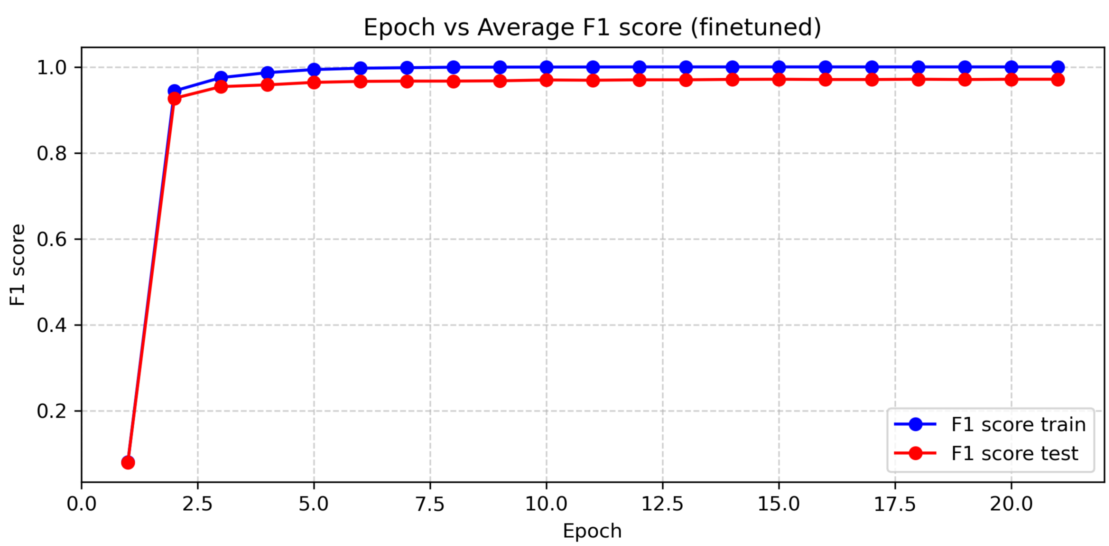


The final F1-score for training data was close to 1. For the test data, it was 0.9707 which shows that the model performs really well.

(f)(i) In this part, the 4 layer architecture is used similar to those in above parts. The model weights for the hidden layers are initialized using the pre-trained model from part (d) which was trained on consonants dataset. 20 epochs were ran for the training. The following learning curve was obtained.

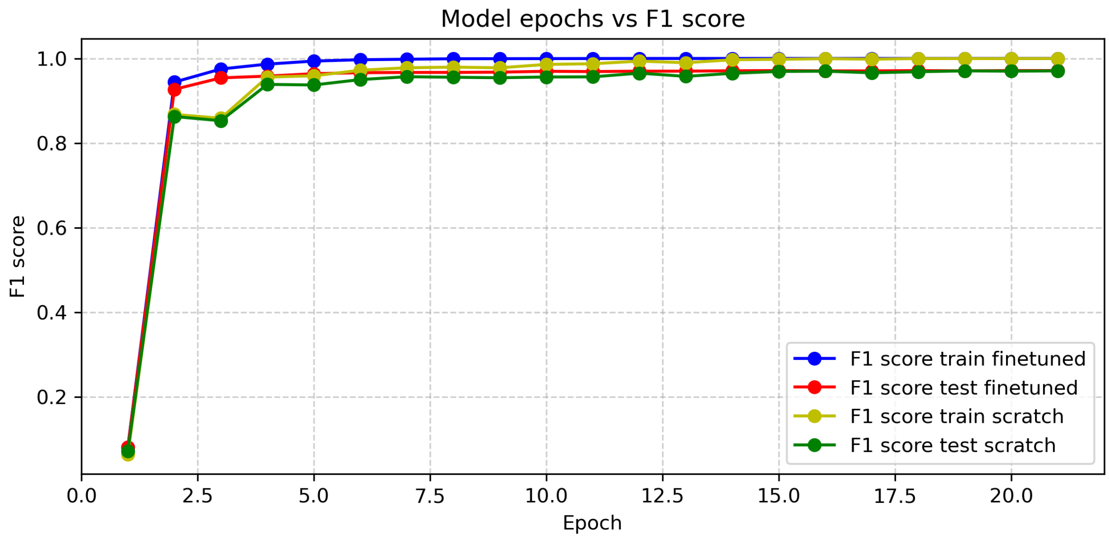


The training loss decreases much more rapidly here than in part (i). The following curve shows how the F1-score changes through the epochs



The final F1-score for training data was close to 1. For the test data, it was 0.9713 which shows that the model performs really well. Also, it performs slightly better than the one in part (f)(i). The F1-score also increases more rapidly in this case.

The following plot shows the F1-scores from both of the above parts in the same plot.



As we can see from the above plot, the F1-score for the model trained from scratch is initially lower than the one which is finetuned but finally curves for the finetuned model and the model trained from scratch become indistinguishable.

Hence using finetuning helps faster convergence and overall faster training of the model.